Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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3.1 Biomedical Information Extraction

3.1.1 Definition

Biomedical Information Extraction (IE) is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-text articles. In this project, IE is treated as the first stage of a larger system in which extracted entities and relations are later assembled into a Knowledge Graph (KG) and validated against biomedical ontologies **[NEED REFERENCE]**.

3.1.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with an analysis by Lu et al. (2011) demonstrating an exponential growth in the number of biomedical publications [1]. This growth is supported by more recent data – a 2024 study mapping over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to a level where the volume of biomedical publications has made it difficult for researchers to keep track of new literature [2]. Literary analysis is not only a volume issue, but also an issue with discovery – log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [3], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

Together, these findings motivate the need for automated IE for two reasons: the breadth of literature is growing faster than it can be analysed [1, 2] and a substantial majority of literature goes unseen that automated extraction and structuring could help mitigate [3]. This motivation supports the choice of PubMed as the source of the contemporary corpus for this project, as its large and continuously updated collection of recent literature reflects the exact setting in which static benchmark-trained pipelines are most likely to be used.

3.1.3 Tasks

Biomedical IE is usually comprised of several sub-tasks: Named Entity Recognition (NER), which identifies and stores spans of text referring to biomedical concepts and assigns them entity types, Relation Extraction (RE), which identifies semantic relationships between recognised entities, and Named Entity Normalisation (NEN), which maps extracted mentions to pre-defined identifiers from an existing ontology [4, 5].

Unlike NER, which only labels a span of text as belonging to a given entity type, NEN maps that span to an established identifier from a pre-defined ontology or vocabulary, storing which specific concept the span is referring to, rather than what kind of concept it is. This resolves terminology ambiguity challenges discussed in the following section.

3.1.4 Challenges

Long or complex sentences, which are common in biomedical abstracts, summarising multiple findings into a single sentence, can cause a significant challenge for these tasks [6]. This is particularly prominent for relation extraction, which typically requires attention over the text's distance between the two entities within a sentence: if two entities are far apart in this text, multiple clauses and mentions of competing entities can make it more difficult to correctly associate the entities with their corresponding relation [7], increasing the risk of both False Negatives (FNs) and False Positives (FPs). Sequential relation extraction models have been shown to have difficulty capturing these long-range dependencies, resulting in poorer performance on longer sentences [8]. The information density of biomedical literature may partly reflect conventions of scientific communication, which emphasise conciseness, clarity and precision of information [9].

Beyond task complexity, biomedical text includes domain-specific challenges that separate it from general IE: dense and evolving terminology [10, 11], and high rates of entity name ambiguity [12], where the same underlying concept is frequently expressed through several distinct strings, such as a drug's brand name versus its generic name, or a gene's full name versus a locally-defined abbreviation within a specific paper, while on the other hand, a single string can refer to different concepts depending on context. As a result, extracted entities referring to the same concept would be treated as distinct, fragmenting information than consolidating it. NEN aims to address this, mapping ambiguous or synonymous mentions onto a shared established identifier. This task is largely important for subsequent KG construction, as graph nodes rely on identifiers to be queryable and to allow separate mentions of the same concept to be merged into a single node **[NEED REFERENCE]**.

In addition to these domain-specific challenges, significant methodological limitations compound the difficulty of RE evaluation. Meaningful comparison between biomedical IE systems requires consistent evaluation frameworks: differences in annotation guidelines, entity and relation identifiers, and train/test splits across benchmark datasets can make performance difficult to compare directly between studies, even when the underlying literature appears homogeneous **[NEED REFERENCE]**. Separately, strong generalisation across literature distributions requires models to perform reliably across heterogeneous literature - literature differing in general publication date, subdomain, terminology, and reporting style from that of what a system was developed and evaluated on **[NEED REFERENCE]**. Benchmark datasets are fixed at the point of annotation, while biomedical literature continues to grow and evolve, therefore, strong performance on a static, in-distribution benchmark therefore does not offer any guarantee of comparable performance on literature published after the benchmark was constructed, or on subdomains that may have been underrepresented within it. This distinction between benchmark-reported and generalised performance is further discussed in Section 3.5, and motivates evaluating the frozen extraction pipeline developed against a separately constructed contemporary corpus.

3.1.5 Traditional Methods and Evaluation

Before generative Large Language Models (LLMs) became viable for biomedical IE, the field progressed through several distinct systems. Initial systems relied on manually-constructed rules and biomedical ontologies, matching text against predefined patterns and dictionaries to identify entities and relations **[NEED REFERENCE]**. These were gradually improved by and progressively replaced by traditional machine learning methods, which used syntactic and contextual features to train classifiers such as support vector machines for NER and RE **[NEED REFERENCE]**. Neural systems were then gradually implemented, initially through architectures such as recurrent and convolutional neural networks trained on word representations and later through transformer-based Pretrained Language Models (PLMs) fine-tuned specifically on biomedical text, such as BioBERT and PubMedBERT **[NEED REFERENCE]**. The progression of pretraining on biomedical corpora and task-specific fine-tuning allowed PLMs to substantially outperform earlier feature-based methods, resulting in them becoming the main extraction-system approach for specialised biomedical IE prior to later prompt-based generative approaches.

Development and evaluation throughout this progression have mainly been driven by benchmark datasets, with models commonly trained and evaluated on fixed, established corpora. However, relying on individual benchmarks introduces comparability and generalisability limitations outlined previously. Recent work has therefore aimed to address these limitations through more consistent evaluation frameworks and broader heterogeneous training data: Sanger et al. (2025) compared multiple PLMs before evaluating knowledge extraction of the best-performing model across five datasets within a consistent evaluation framework

[13]. Similarly, Lai et al. (2023) introduced BioREx, which consolidates heterogeneous biomedical RE datasets to support broader training and improve generalisation [14].

Both of these approaches, however, rely on PLMs tuned on labelled benchmark data, with performance assessed against fixed, in-distribution, annotated datasets. This highlights an important methodological limitation for practical application, where extraction systems must perform robustly on dynamic, contemporary literature, rather than a static benchmark corpus.

3.2 Large Language Models for Biomedical Information Extraction

3.2.1 Transition from Specialised Models to LLMs

More recently, LLMs have driven a broader shift in knowledge extraction and KG construction from rule-based and module pipelines towards generative and adaptive frameworks [15]. Early evidence supports competitive extraction performance with Zhang et al. (2024) reporting an F1 score of 0.84 for GPT-4 on biomedical RE for the Genetic Association Database (GAD) benchmark dataset, with GPT, BioBERT, and PubMedBERT performing comparably on the GAD dataset under certain extraction system frameworks [16]. However, it is important to note that GPT-3.5 and BioBERT v1.1 performed considerably worse than PubMedBERT on the masked ChemProt dataset, suggesting that relative model performance may vary substantially across biomedical RE tasks and datasets.

Additionally, performance gains from LLMs are not limited to biomedicine. Evaluations of LLMs for KG construction have observed an F1 score of 99.35% for Meta's LLaMA 3 model on OpenStack system log data [17], indicating extraction capability is not domain-specific.

This body of evidence motivates the pipeline design adopted in this project: an LLM-based extraction approach using structured JSON outputs and few-shot prompting, developed on an existing benchmark, BioRED [18], before evaluation on a newly-assembled contemporary corpus.

3.2.2 Prompt-Based and Few-Shot Extraction

Unlike specialised pretrained language models such as BioBERT and PubMedBERT, which require task-specific fine-tuning on labelled data to perform NER or RE, generative LLMs can perform biomedical IE directly through prompting, without any parameter tuning [19]. This approach can allow flexibility through natural-language instructions, rather than requiring additional model training. As a result, a single general-purpose model can potentially be adapted to different entity types, relation schemas, and extraction tasks through only prompt refinement

[NEED REFERENCE].

Several approaches can be implemented: zero-shot prompting - where the model is given only a natural-language description of the task, few-shot prompting - where the prompt includes structured examples of exemplary outputs corresponding to given inputs, and schema-guided instruction, where the prompt specifies the exact entity and relation types that the model should extract [20, 21, 22].

The extent to which an LLM is able to compete with fine-tuned extraction systems depends heavily on how the prompt itself is constructed. Recent comparisons between zero-shot and few-shot prompting on biomedical IE tasks suggest that including examples in the prompt can improve extraction quality, as can annotation guidelines or provided schema within the prompt itself [21]. However, these gains are not consistent across biomedical IE tasks: Zhang et al. (2024) observed only limited improvements from one-shot prompting on some datasets, with performance decreasing on the ChemProt dataset. Given ChemProt's more complex relation schema [23], these findings suggest that the effectiveness of few-shot prompting may depend on task complexity and how well the selected examples represent the target annotations **[NEED REFERENCE]**. This prompt-dependency is important for this project specifically as it provides a basis for developing and iterating on multiple Extraction Pipeline Iterations (EPI) during BioRED-based development, rather than treating prompt design as a single, fixed choice that remains unexamined.

This has a direct implication for how prompts are constructed in this project. Recent work testing LLM annotation directly against BioRED and other biomedical benchmark datasets found that providing the benchmark's own entity schema and annotation guidelines within the prompt substantially improved alignment with the benchmark's annotations compared to a prompt including only entity schema, excluding specific guidelines, across every model and dataset tested [24]. The study's authors discuss how this is not because of a failure of the underlying model's biomedical knowledge, but misalignment between the model's interpretation of the extraction task and the conventions that the benchmark's annotations follow [24]. This distinction is important for how benchmark-oriented prompting should be understood: including a benchmark's schema into a prompt is a matter of methodology, defining what the extraction system is instructed to extract, rather than overfitting to the benchmark. This motivates tuning each EPI developed in this project towards BioRED's specific entity and relation schema and guidelines, rather than relying on a generic extraction prompt as it aligns the pipeline's output with the specific ground truth it is developed against.

However, this choice also defined the scope of what Research Question 2 (RQ2) is able to evaluate. A pipeline based on a prompt that is tailored to a benchmark's schema it optimised to perform the specific extraction task the benchmark defines, rather than a domain-level general extraction task. Evaluating this project's final frozen pipeline against contemporary literature therefore investigates whether its fixed, BioRED-based task specification transfers across differing text distributions, not whether the extraction system generalises to other unseen annotation schemas. This project is therefore scoped as an investigation of this specific research question, rather than the development of a general-purpose, production-oriented extraction system. BioRED-geared prompting is therefore treated as the fixed task configuration that is used to answer RQ2.

3.2.3 Performance and Limitations

Despite this flexibility, LLM-based extraction introduces failures that are usually absent from, or less significant in fine-tuned supervised models **[NEED REFERENCE]**. The most widely discussed is drawing from domain knowledge: LLMs may produce extracted entities or relations that are plausible but not present in the supplied text [22]. This is particularly important for biomedical IE, where an extracted relation is only useful if it is grounded in the given literature, rather than drawn from existing background knowledge.

Problems also exist with LLM consistency and control: LLM outputs can vary across repeated runs on the same input [25], and extraction quality has been shown to be sensitive to changes in prompt wording or the specific examples chosen for few-shot prompting [16, 21]. Models may also fail to strictly follow a predefined relation schema, extracting relation types outside the intended ontology or omitting required output fields, which can complicate downstream pipelines that assume a fixed schema [25]. Together, this behaviour may create a precision-recall trade-off where more exhaustive extraction increases recall at the cost of precision, while more conservative prompting tends to do the opposite.

3.2.4 Structured Output

An important design consideration is the format in which extracted information is returned. Limiting LLM output to a structured, machine-readable format, such as JSON conforming to a predefined schema, rather than unstructured text generation is intended to make extraction results easily usable in downstream processing without requiring separate parsing steps [22]. For a pipeline that extracts entities and relations that are subsequently assembled into a KG (Section 3.3), this is an important methodological choice: KG construction requires each extracted entity and relation to be reliably identifiable as a discrete unit whereas unstructured free-text output requires an additional extraction from the output extractions. Structured output generation is therefore treated in this project as a component of the extraction pipeline itself, rather than a post-processing step.

3.2.5 Implications for Biomedical Information Extraction

In summation, the current literature provides a more nuanced view than the conclusion that LLMs perform interchangeably with specialised biomedical extraction models. LLMs reduce the dependence on task- and subdomain-specific fine-tuning and offer flexible, adaptable extraction across entity and relation types, which is an important practical advantage over supervised PLM pipelines discussed in Section 3.2.1. However, this flexibility comes with significant caveats: extraction performance that is sensitive to prompt construction and appears to vary across datasets and specific tasks while the generative aspect of these models comes with reliability concerns such generation of information not grounded in the source text, variable outputs, and schema non-compliance. For a pipeline intended to generalise from a benchmark corpus to unseen contemporary literature, this distinction is important: benchmark-evaluated precision and recall capture extraction accuracy on in-distribution text, but do not directly capture whether a frozen prompt-based pipeline remains robust when applied to literature it was not developed against. This motivates evaluating not only whether extraction quality transfers from BioRED to contemporary PubMed abstracts, but whether the specific points of failure discussed become more or less significant during generalisation.

3.3 Knowledge Graph Construction Using LLMs

3.3.1 Knowledge Graph Representation

A KG represents information as a set of entities, or nodes, connected by relations, or edges, with both nodes and edges, sometimes including additional information such as a confidence score, or source identifier **[NEED REFERENCE]**. The fundamental unit of a KG is the triple, which represents a single relationship between two entities. For example, the finding that warfarin is associated with intracranial haemorrhage can be represented as the triple:

$$(\text{Warfarin}, \text{Association}, \text{Intracranial Haemorrhage}).$$

A KG is fundamentally a collection of triples, with shared entities linking separate triples into a connected structure rather than a set of isolated relations.

This structure is well suited to biomedical information where individual findings are often reported in isolation within a single abstract but gain insight when connected across the literature. Representing extracted findings as a knowledge graph connects mentioned of the same entity across different papers into a single queryable structure. This can reveal clusters of relations and indirect connections that would not be apparent from a single abstract in isolation. For example, identifying an indirect association between two entities that are never mentioned together in the same source text, but are connected to each other through a mutual entity **[NEED REFERENCE]**.

This representation maps naturally onto the information extraction tasks introduced in Section 3.1: NER identifies the spans of text that become candidate nodes in the graph and NEN maps those candidate nodes to shared predefined identifiers, ensuring that semantically separate mentions of the same concept are consolidated into a single node rather than represented as duplicates. Finally, RE identifies the edges connecting these nodes. The insights KGs provide are precisely why NEN is a key component of the extraction pipeline: if duplicate mentions of a single concept were not resolved, biomedical concepts would become fragmented across several nodes, undermining the purpose that KGs are intended to provide.

3.3.2 LLM-Based Knowledge Graph Construction

Section 3.2.1 outlined how the recent LLM-driven work represents a broader shift in knowledge extraction from rule-based and modular pipelines towards adaptive generative frameworks [15]. In the context of KG construction, this shift changes the overall architecture of the entire extraction-to-graph system. A traditional, modular architecture treats NEN, NER, and RE as separate stages, each performed by a distinct model or tool, before the resulting entities and relations are assembled into a graph **[NEED REFERENCE]**. Although the exact ordering of these steps may differ between models, an example module workflow is:

$$\text{Text} \rightarrow \text{RE} \rightarrow \text{NER} \rightarrow \text{NEN} \rightarrow \text{KG } \textbf{[NEED REFERENCE]}.$$

An LLM-driven approach instead allows a single model to conduct RE and NER directly from text within one pipeline step, with a separate subsequent step conducting NEN before KG construction:

$$\text{Text} \rightarrow \boxed{\text{LLM joint NER + RE}} \rightarrow \text{NEN} \rightarrow \text{KG}.$$

This consolidation is possible as an LLM with a single adequate prompt can both recognise entity spans and the relations between them in text. Recent work evaluating end-to-end biomedical relation extraction using LLMs demonstrates the potential of joint-extraction [22] (?) and multi-extraction prompting has also been shown to support this within a single framework without requiring additional fine-tuning [25].

Furthermore, because the LLM's output is constrained to an easily-parsable schema, rather than free text, the extracted entities and relations are already in a form that does not require extra parsing or interpretation to be mapped onto graph triples. This is what allows the LLM-driven architecture to abstract several traditionally separate pipeline stages into fewer steps.

It is worth noting that LLMs can support KG construction beyond direct entity and relation extraction, including schema and ontology generation or refinement [15] **[NEED REFERENCE]**. This project primarily uses an LLM-based approach for structured information extraction, while also implementing LLM-assisted error analysis during iterative development of the extraction system. The graph schema and ontology itself, however, remain predefined, rather than being generated or refined by the LLM.

3.3.3 Ontology Integration and Entity Normalisation

Within KG construction, NEN provides the connection between entity mentions extracted from free text and established biomedical concepts. As discussed in 3.1.4, relying on separate nodes formed by unmapped string would fragment mentions of the same biomedical concepts across the graph. Mapping these mentions to ontology identifiers instead allows references to the same underlying concepts to be consolidated.

Biomedical ontologies and vocabularies address this by providing a fixed set of biomedical concepts, each associated with an established identifier, that extracted mentions can be mapped to **[NEED REFERENCE]**. NEN therefore acts as the step that allows a free-text span to be represented by an identifiable graph node.

Beyond identifier assignment, the ontology can also constrain the KG more broadly, restricting the extracted entities and relations to respective predefined, established sets **[NEED REFERENCE]**. This introduces a distinction between normalisation and ontology-based validation. While NEN maps an extracted span of text to a consolidated, established concept, validation can assess whether extracted entities and relations align with existing ontologies. Ontology integration can therefore support the consolidation and validation of the resulting KG **[NEED REFERENCE]**.

3.3.4 Challenges and Quality of LLM-Constructed Knowledge Graphs

Because a KG is assembled directly from the outputs of the extraction pipeline, errors made in the NER, RE, and NEN stages are not isolated from other stages, but instead pass into the resulting graph. For example, a false-positive entity or relation will become an incorrect node or edge, while false negatives would become missing edges. False-positive relations may be the more detrimental of the two errors, as the graph explicitly states them as knowledge rather than just excluding them **[NEED REFERENCE]**. Errors at the NEN stage, discussed in Section 3.3.3, have a similar effect: false-negative normalisation results in mentions of the same concept separated across duplicate nodes, while false-positive normalisation consolidates separate concepts into a single node **[NEED REFERENCE]**.

Because of the connected nature of steps in an extraction pipeline system, KG quality is dependent on the quality of the original extraction. A KG can also be evaluated at the graph level for properties such as consistency, completeness, duplication, and compliance with its ontology or schema **[NEED REFERENCE]**.

3.4 Benchmark Datasets and Evaluation

3.4.1 Biomedical Information Extraction Benchmark Datasets

Progress in biomedical IE, has largely been measured against manually annotated benchmark corpora, which provide the ground truth against which NER and RE systems are trained and evaluated **[NEED REFERENCE]**. As annotation is costly and requires domain expertise, researchers typically constrain their annotations to particular biomedical entities, relations, or domains **[NEED REFERENCE]**, targeting a specific set of entity and relation types, rather than biomedical text in general **[NEED REFERENCE]**. Because of this, different benchmarks are not directly interchangeable as an extraction system's reported performance reflects the entities, relations, and schema of the specific benchmark it was evaluated on, rather than a measure of their general biomedical extraction ability. This benchmark heterogeneity was discussed in Section 3.2.1, where performance was shown to vary significantly between the GAD and ChemProt datasets.

This heterogeneity is stated briefly in table **[NEED REFERENCE]**, comparing the domain and entity/relation scopes of three benchmark datasets previously mentioned in this report.

Table 3.1: Comparison of biomedical relation extraction benchmark datasets.

Dataset	Subdomain	Entity/Relation Type	Project Relevance
GAD [26]	Genetic association	Gene-disease association pairs: single binary relation type	Narrow, single-relation benchmark
ChemProt [23]	Chemical-protein interaction	Chemical and protein entities: multiple interaction subtypes	Benchmark with a more complex relation schema
BioRED [18]	General biomedical abstracts	Multiple entity types (e.g., gene, disease, chemical, species) and multiple relation types	Broader biomedical coverage

This comparison demonstrates that the interpretation of a strongly-performing extraction system is dependent on the dataset it is built on: a model exhibiting strong results on the GAD dataset demonstrates a narrower scope of extraction strength than that of the BioRED dataset as the two differ substantially in terms of how many entity and relation types must be extracted and, therefore, understanding of context from biomedical literature. This difference motivates the choice of BioRED as the benchmark dataset used to develop the extraction pipeline in this project as a benchmark covering multiple entity and relation types is more representative of the broader task of general-purpose KG construction.

3.4.2 BioRED

BioRED, introduced in Section 3.2.1, is a manually annotated corpus of 600 PubMed abstracts, constructed to support document-level, multi-entry, multi-relation biomedical relation extraction **[NEED REFERENCE]**.

The corpus comprises six entity types: chemical entity (`ChemicalEntity`), disease or phenotypic feature (`DiseaseOrPhenotypicFeature`), gene or gene product (`GeneOrGeneProduct`), sequence variant (`SequenceVariant`), organism taxon (`OrganismTaxon`), and cell line `CellLine`. Eight relationship types connect these entities: association (`Association`), bind (`Bind`), comparison (`Comparison`), conversion (`Conversion`), cotreatment (`Cotreatment`), drug interaction (`Drug_Interaction`), negative correlation (`Negative_Correlation`) and positive correlation (`Positive_Correlation`) [18].

In total, the corpus contains 20,419 entity annotations, mapped to 3,869 identifiers **[CHECK INFO]**, where, each entity mention is normalised to an established identifier in the annotation itself [18]. Relations are also labelled as describing either a new finding or existing background knowledge, allowing systems to distinguish novel findings from existing background information. As with the other benchmark corpora discussing in this review, BioRED is split into training, development, and test splits, enabling consistent comparison between systems evaluated on the dataset [18].

Because of these characteristics of the BioRED dataset, BioRED is well suited to the aims of this project. While other commonly-used biomedical RE benchmark datasets are constructed around a single entity or relation type, BioRED's wide scope of multiple entity types and relations and existing entity normalisation align closely with this project's general-purpose extraction task required for KG construction. This scope is the main motivator for its selection as the benchmark dataset used to iteratively develop an extraction pipeline in this project, ahead of evaluation against contemporary literature.

However, BioRED remains as fixed, manually annotated, benchmark-distributed corpus where annotations reflect the 600 abstracts sampled during construction and its entity and relation schema remains fixed rather than reflective of the full scope of biomedical concepts and relationships in contemporary literature. Because of this, while BioRED provides reliable ground truth for developing and evaluating an extraction pipeline, performance on the dataset does not necessarily reflect performance on literature published after the benchmark or on subdomains it was not designed to represent. BioRED is therefore treated in this project as a source of ground truth for extraction pipeline development, rather than as evidence of generalised extraction performance.

3.4.3 Extraction-Level Evaluation

Comparing biomedical IE systems, whether between the benchmark datasets in Section 3.4.1 or traditional and LLM-based approaches discussed in Sections 3.1.5 and 3.2, requires a consistent way of quantifying extraction quality. This is typically done by comparing a system's predicted entities or relations against a benchmark's manually annotated ground truth, classifying each prediction as a True Positive (TP), a correct prediction matching an annotation, a False Positive (FP), a predicted instance with no associated annotation, or a False Negative (FN), an annotation that the system failed to extract **[NEED REFERENCE]**.

Three derived metrics are then calculated from these counts: precision, the proportion of all predicted extractions that are correct, recall, the proportion of all ground truth annotations that were correctly identified from the source text, and F1 score, the harmonic mean of precision and recall, commonly reported as a standalone summary figure, previously mentioned in Section 3.2.1 **[NEED REFERENCE]**.

Accuracy measures the proportion of predictions that are correct:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

However, biomedical extraction tasks usually involve many possible entity or relation candidates where true negatives can dominate. Accuracy can therefore be misleadingly high when a system extracts very little. In contrast, precision, recall, and F1 better represent the balance between over- and under-extraction discussed in Section 3.2.3 **[NEED REFERENCE]**. This framework applies both at the entity and relation levels. At the entity level, predictions are evaluated as span-type pairs, whereas relation-level predictions are made up of a pair of entities and the relation connecting them. Relation-level performance therefore also depends on the correctness of the underlying entity predictions **[NEED REFERENCE]**.

Additionally, the matching criteria used to determine whether an extracted entity or relation is correct should be noted. Under strict matching, a predicted entity span, type, or relation must exactly match an annotation to be counted as a true positive, with any boundary or string mismatch classifying the extraction as an FP and the ground truth annotation as an FN **[NEED REFERENCE]**. Using relaxed matching, some variation in the span, type, or relation is allowed, given the extraction still identifies the correct underlying concept **[NEED REFERENCE]**.

This distinction is imperative for LLM-based extraction specifically. As LLM outputs are generated rather than selected from a fixed set of labels, a predicted entity or relation may be semantically equivalent to an annotated instance while differing in exact expression – whether that is variation in exact string or its span boundary. Strict matching incorrectly classifies this variation as a genuine false negative, even when the underlying concept is correct. This motivates using a more relaxed or flexible matching approach when evaluating LLM-based extractions, illustrating the general limitations of strict matching for generative extractions. This project introduces supplementary evaluation approaches given the generative, variable nature of LLM-based extractions. The implementation of this approach is discussed in Section **[MORE DEPTH]**.

3.4.4 Limitations of Benchmark Evaluation

There are three distinct limitations that should be noted for benchmark evaluation, appearing at different parts of the evaluation pipeline.

The first is the dependence on a system's benchmark ground truth. Precision, recall, and F1 only reflect extraction performance relative the specific benchmark a system was built on as these metrics are only calculated directly against a fixed set of manual annotations. A predicted entity or relation that is correct in the broader scope of biomedicine can still be classified as an error if it does not meet the benchmark's specific annotation guidelines, entity boundaries, or relation definitions **[NEED REFERENCE]**. Performance figures therefore reflect a system's alignment with its benchmark's specific set of annotations, rather than its absolute extraction performance.

Secondly, building on the first limitation, as different benchmarks define different annotation schemas, F1 scores from systems build on different benchmark datasets are not necessarily directly comparable. A high F1 score on one benchmark and a lower F1 on another may not reflect a difference in extraction capability, rather a difference in schema complexity or annotation strictness between them. This has been observed previously in this literature review in Section 3.2.1 with the GAD/ChemProt datasets.

The third limitation, and most relevant to this project, is the external validity. Strong performance on a benchmark demonstrates only that a system performs well against its benchmark's specific collection of text and paper-level entities and relations. It does not inherently demonstrate that the same performance will hold against literature published after the benchmark was established or against subdomains and academic reporting practices that the benchmark did not sample.

In summation, these limitations do not invalidate the value of systems built upon benchmark datasets, rather their recognition helps to establish their scope. Benchmark evaluation remains essential as it provides controlled ground truth on which an extraction pipeline can be developed on. However, a system developed on a benchmark dataset cannot, by itself, demonstrate its generalisation to a separate distribution of literature. Addressing that gap is the motivation for this project's second research question.

3.5 Generalisation to Contemporary Biomedical Literature

3.5.1 Distribution Shift in Biomedical Literature

The limitations of benchmark evaluation discussed in Section 3.4.4 rely on the underlying assumption that a benchmark's fixed sample of text is representative of the literature a system will later be applied to. This assumption is important enough to warrant exploration as it provides the premise that the generalisation question this project investigates depends upon.

Biomedical literature is not static [2, 1]: new terminology and abbreviations become more commonly used, newly discovered entities and concepts are discovered, research focuses may shift over time, and reporting conventions may evolve **[NEED REFERENCE]**. This difference in the data a model or system is trained or tuned on and the data it encounters in real-world practice is generally called a distribution shift [27] and has been briefly discussed previously in Sections 3.1.4 and 3.4.4 in the observation that benchmark datasets are fixed while the literature they are intended to represent continues to grow.

A benchmark such as BioRED captures only a historical sample of biomedical literature, comprising abstracts available at the time of construction. As biomedical literature continues to grow evolve, later publications may shift away from the distribution that the benchmark represents, even where the general extraction task of identifying entities and relations from the biomedical abstracts remains the same. It should be noted that a distribution shift does not imply that a benchmark becomes invalid as a ground truth, but that it cannot be assumed that its representativeness of contemporary literature remains constant.

Literature exploring evidence for this caveat in the biomedical domain is relatively limited, as most existing work on distribution shifts over time has focussed more on general literature, rather than biomedical corpora specifically **[NEED REFERENCE]**. However, a recent study by Liu et al. (2025) measuring temporal effects across biomedical language models found that performance fell when models trained on data from one time period were evaluated on data from a later period, with the extent to which performance degraded depending on the specific task and the magnitude of the temporal shift [28]. Furthermore, NER exhibited greater sensitivity to shifts over other extraction tasks. This provides direct yet still scarce evidence that the distribution shift assumption associated with benchmark-based development is an important consideration within biomedical IE.

Together, this raises the uncertainty about whether performance established on BioRED will be maintained when the same extraction pipeline is applied to newer biomedical literature drawn from a newer distribution.

3.5.2 Generalisation of Biomedical Extraction Systems

The discussion of generalisation ability, or lack thereof, in Section 3.5.1 has established why contemporary biomedical literature may differ from a fixed benchmark's distribution. This section (Section 3.5.2) builds on the previous discussion to provide deeper review of how extraction performance differs when a system is evaluated outside the distribution it was developed on.

As discussed previously, BioREx, introduced in Section 3.1.5, was proposed specifically to improve generalisation through heterogeneous training across multiple biomedical RE datasets, rather than relying on a single benchmark's training data. Similarly, Sanger et al.'s 2025 evaluation of a single best-performing model across five different datasets was motivated by the same general problem: performance obtained on one biomedical dataset does not directly transfer to another [13]. Both examples indicate that cross-dataset generalisation in biomedical RE is a significant problem requiring explicit attention.

Further evidence comes from literature evaluating the same models across multiple biomedical relation classification datasets side by side. Purpura et al. (2024) compared fine-tuned Bidirectional Encoder Representations from

Transformers (BERT)-based models (DistilBERT and PubMedBERT) against zero- and few-shot LLMs (Flan-UL2 and Llama 2) on three distinct biomedical relation classification datasets: GAD, the Compound-Protein Interaction (CPI) dataset, and a social-determinants-of-heath dataset [29]. Performance varied significantly between all three dataset for each model, with F1 scores on the GAD dataset substantially lower than those of the CPI dataset for the same models. This further supports the evidence from Section 3.2.1 that difference in extraction system performance exist between datasets, however, Purpura et al. (2024) demonstrates the same effect across all combinations of several different models and datasets.

Together, this evidence provides the basis of a more cautious view than the claim that LLMs generalise better. LLMs provide a suitable extraction system architecture for transferable tasks, given their ability to perform NER and RE without task-specific fine-tuning as discussed in Section 3.2.1. However, the existing literature does not conclude whether this flexibility translates into stable performance when a pipeline developed on one benchmark is applied to an unseen distribution. Specifically, whether an LLM-based extraction pipeline developed on BioRED maintains its performance when applied to newly assembled contemporary biomedical literature remains relatively unclear.

3.5.3 Research Gap and Implications

All previous sections of this literature review have established several main points of discussion: that automated biomedical IE is necessary given the scale of biomedical literature, that specialised PLMs achieve strong performance when fine-tuned on benchmark data, that LLMs offer easily adaptable extraction without task-specific fine-tuning while being sensitive to prompt, task, and dataset choice, that extraction errors can spread forward in into any KGs built subsequently, and that benchmark evaluation provides controlled, reliable ground truth but cannot by itself validate its performance on out-of-distribution data.

Together, these findings provide the basis for RQ2. Existing literature provides evidence of both distribution shift and variation in extraction performance across datasets, but does not, however, directly test the validity of LLM-based-extraction benchmark-to-contemporary transfer examined in this project. Specifically, the difference in performance observed by Liu et al. (2025) was measured across fine-tuned biomedical PLMs, rather than prompt-based LLMs operating without task-specific tuning, and Purpura et al.'s cross-dataset comparison, evaluates transfer between existing, already-annotated benchmarks rather than to independently sampled, contemporary literature. Whether an LLM-based extraction pipeline, developed and iterated against a fixed benchmark dataset, retains its extraction quality when applied unchanged to newly published biomedical literature therefore remains unclear.

This project addresses that gap directly. An LLM-based extraction pipeline is developed and iterated against BioRED, before being frozen and applied without further refinement to an independently assembled contemporary corpus, allowing its benchmark-level performance to be evaluated against a distribution it was not developed on.

Chapter 4

Methodology

4.1 Research Design

This study uses an experimental pipeline development design in which an LLM-based biomedical information extraction system is developed iteratively against a fixed benchmark corpus before being frozen and evaluated on independently sampled, contemporary biomedical literature. The study consists of four main stages: development of the extraction pipeline against BioRED, selection of a pipeline iteration, evaluation of the chosen iteration, and construction and evaluation of a knowledge graph from the resulting contemporary extractions.

The main methodological idea of this design was the separation between development and evaluation. BioRED served as a basis for training, development, and test splits where this project's iterative pipeline development, made up of a set of Extraction Pipeline Iterations (EPIs), was conducted solely against the training and development splits. Each EPI's prompt design and configuration was based on the previous EPI's performance and optional further error analysis — this approach began with an initial configuration (EPI 001), discussed further in Section 4.5, that established a baseline extraction pipeline using generic biomedical extraction instructions without BioRED-specific schema guidance. Pipeline development continued against the BioRED training split until performance plateaued, at which point the EPI configurations were frozen and no further prompt refinement or schema adjustment was performed. The frozen iterations were then evaluated against the BioRED development split, where their performance was compared with one another and the best performing EPI was selected. This distinction allows subsequent evaluation to be treated as a measure of the pipeline's performance, rather than further development.

The final EPI was then evaluated under two distinct stages. The first is benchmark evaluation, which measures the final EPI's extraction performance against BiORED's held-out ground truth, establishing its performance within the distribution it was developed on. The second stage was generalisation evaluation, which applies the same pipeline to an independently assembled corpus of contemporary PubMed literature that the pipeline was not iteratively developed on. As the pipeline was not adapted to perform specifically on the contemporary corpus, any significant degradation in performance between them can be investigated as evidence of reduced generalisation to the contemporary corpus. Following extraction and normalisation on the contemporary corpus, the extracted entities and relations were assembled into a KG.

4.2 Datasets

4.2.1 BioRED Benchmark Corpus

The original BioRED corpus, as described in Section 3.4.2, was used as the benchmark corpus for extraction pipeline development and evaluation in this project. The corpus consists of 600 manually annotated PubMed abstracts, assembled and released by the National Center for Biotechnology Information (NCBI) and was obtained directly from the dataset's public GitHub repository. The BioRED-BC8 extension was also considered as it provides a more recent expert-annotated evaluation set: however, the original BioRED corpus was retained as the controlled benchmark as the study separately evaluates generalisation using literature independently retrieved through the PubMed Application Programming Interface (API), better reflecting the intended deployment pathway. BioRED-BC8 therefore represents a valuable alternative external evaluation dataset for future work.

The original version of the BioRED dataset divides 600 abstracts into a training split of 400 abstracts, a development split of 100 abstracts, and a test split of 100 abstracts. Each split was used for a distinct role in this study. The training split was used throughout iterative EPI development, providing the abstracts observed by the LLM in which all EPI configurations were developed on, the development split was then used to evaluate all frozen EPI configurations against one another, and the test spit was used exclusively for the final benchmark evaluation of the best-performing pipeline configuration and played no role in either pipeline development or EPI selection.

For each abstract, this project used the associated PubMed Identifier (PMID), title, and abstract text as the input passed to the extraction pipeline, together with the corresponding gold-standard annotations used as ground truth. These annotations consisted of each entity's type and text span and each relation's type. The following table summarises how each component of the BioRED dataset was used in this study:

Table 4.1: BioRED Split Summary

BioRED Component	No. of Abstracts	Purpose
Training spit	400	EPI development and performance estimate
Development split	100	EPI selection (held-out within BioRED)
Test split	100	Final benchmark evaluation

It should be noted that EPI refinement was conducted using a fixed 35-abstract subsample of the BioRED training split, with each final EPI then evaluated across the full 400-abstract training split. Section **[NEED REFERENCE]** discusses this methodological decision further.

BioRED's train, development, and test splits were each retrieved from the official BioRED GitHub repository as separate BioC-XML files. This process is discussed in Section 4.2.3.

4.2.2 Contemporary Biomedical Corpus

An independently-assembled corpus of contemporary biomedical PubMed abstracts was constructed as the external evaluation set used to investigate generalisation to out-of-distribution data, as mentioned in Section 3.5. This Contemporary Biomedical Corpus (CBC) included abstracts from papers published across 2024, 2025, and 2026 making up 282 total abstracts after duplication removal by PMID and removal of entries with missing abstract texts. It should be explicitly noted that this corpus was assembled entirely independently of BioRED and played no role in EPI development or selection.

"Contemporary biomedical literature" was defined through a fixed set of either PubMed research queries, designed to give broad coverage of the wider field of general biomedicine, rather than a single subdomain. Four of the eight related to individual biomedical entity categories, generally mirroring BioRED's own entity type (gene/protein, disease/disorder, drug/chemical, and variant/mutation), while the remaining four combined two entity categories within a single query (gene and disease, drug and disease, gene and drug, variant and disease). This query configuration was chosen to feasibly maximise the likelihood of retrieving abstracts that include entity relationships, rather than only single entity mentions. Each of the eight queries were combined with a publication year filter for all three publication years, giving 24 distinct query-year combinations with 15 articles requested per combination. The table below summarises each query category and associated term.

Table 4.2: Contemporary Biomedical Corpus Retrieval Query Summary

Query category	Query terms
Gene/protein entity mention	`"gene" OR "protein"`
Disease/disorder entity mention	`"disease" OR "disorder"`
Drug/chemical entity mention	`"drug" OR "chemical"`
Variant/mutation entity mention	`"variant" OR "mutation"`
Gene-disease relation	`"gene" AND "disease"`
Drug-disease relation	`"drug" AND "disease"`
Gene-drug relation	`"gene" AND "drug"`
Variant-disease relation	`"variant" AND "disease"`

Abstracts were retrieved from PubMed using the NCBI Entrez E-utilities, accessed through Biopython's `Bio.Entrez` module under a registered email address, as required by NCBI's usage policy. For each of the 24 query-year combinations an `esearch` request was used to return 15 matching PMIDs, article title, abstract text, journal title, and publication year were extracted. Author names, Digital Object Identifiers (DOIs), and keywords were not stored.

Across all query-year combinations, up to 360 records were requested in total: 15 results per query-year combination. After consolidating all returned results, duplicates were removed where PMIDs matched, along with any records with absent abstract text, leaving a final corpus of 282 abstracts. Beyond publication year, no sampling constraints were applied at this stage. A subset of the CBC was selected for manual annotation, in order to provide ground-truth entity and relation labels which the frozen extraction pipeline could be evaluated against. This subset comprised 50 randomly sampled abstracts from the 282 total entries. The annotation procedure and resulting metrics are discussed in Section 4.6.4.

4.2.3 Data Preparation

Although BioRED and the CBC were retrieved from different sources and through different processes, both required conversion into a structured, tabular representation before use by the rest of the pipeline. `pandas` DataFrames were used for both datasets, allowing later extraction and evaluation to be conducted easily over a tabular data, regardless of which dataset was used. BioRED was retrieved as raw BioC-XML and was parsed directly using `xmltodict`, whereas the CBC was retrieved via Biopython's `Bio.Entrez` module, which returns PubMed records already parsed into Python dictionaries, requiring no further XML-parsing step.

For BioRED, each of the three split's files was parsed with `xmltodict` into a Python dictionary before being converted into three separate `pandas` DataFrames per split: a DataFrame containing each abstract's PMID, title, and abstract text, an entity DataFrame containing one row per annotated entity mention, including its text span, entity type, and normalised identifier, and finally an entity relation DataFrame containing one row per annotated entity relation, including normalised identifiers of the two entities and their relation type.

For the CBC, the entire returned dictionary was iterated over to extract each paper's PMID, title, abstract text, and publication year. Where a paper had its abstract split into multiple segments, all segments were concatenated into a single string and duplicate records.

Relatively little additional preprocessing was conducted beyond this stage. For BioRED, no records were excluded during parsing and for the CBC, no records were excluded beyond removal of duplicates and papers that had missing abstracts. For both corpora, all abstracts and corresponding title text string remained untouched as retrieved as the extraction pipeline provides abstract and title text directly to the LLM as natural language, rather than preprocessed text. The output of this stage was therefore a set of standardised, pipeline-ready CSV files.

4.3 System Architecture

The processed datasets discussed in Section 4.2.3 formed the inputs to the extraction stage of the system. For BioRED, the resulting extractions were used for benchmark evaluation, whereas CBC predictions were subsequently passed through normalisation, validation, and KG construction. For each abstract, its title and text were passed to an LLM-based extraction step, which performed NER and RE jointly rather than as separate LLM API calls, returning structured entity and relation predictions. These predictions were parsed into the same tabular format used for BioRED's own annotations, allowing predicted and ground-truth extractions to be compared directly for subsequent evaluation. Figure **[NEED REFERENCE]** below summarises this data flow across the implemented pipeline architecture.

[MORE DEPTH]Extracted entity mentions then underwent entity normalisation, mapping each mention to its corresponding Concept Unique Identifier (CUI) in the Unified Medical Language System (UMLS), using SapBERT [30] to encode each mention and retrieve its nearest neighbour among UMLS concept vector embeddings by cosine similarity, before undergoing ontology-based validation, described in Section 4.7.2. The resulting normalised, validated entities and relations were subsequently converted into nodes and edges and assembled into the final KG.

The same extraction architecture was used across both BioRED and the CBC, differing only in the input data supplied and presence of gold-standard annotations, with BioRED additionally providing ground truth against which extraction-level performance could be measured, and the CBC providing only a manually annotated subset for this purpose, as described in Section 4.2.2. Although the pipeline's specific configuration changed across the EPI development process, the overall architecture described above remained consistent throughout.

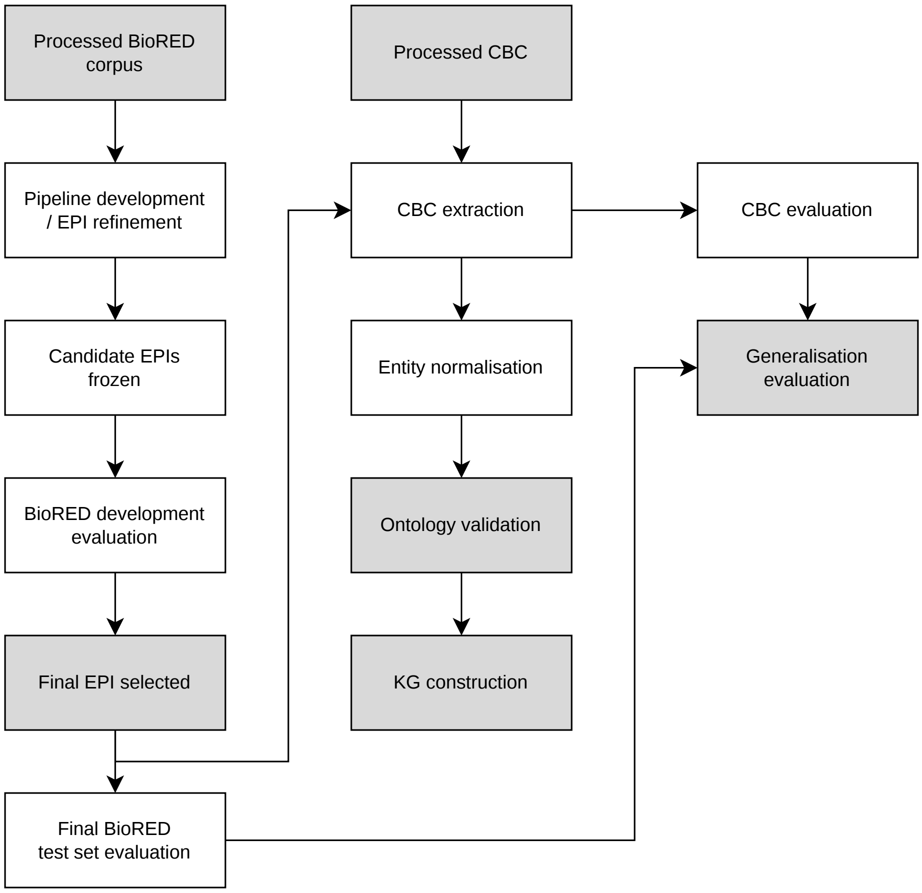

Figure 4.1: System architecture and data flow, from prepared corpus input through extraction, normalisation, and KG construction, to the three evaluation branches described in Section **??**. Grey shading highlights key inputs, final stages, and outputs.

4.4 Information Extraction Pipeline

Prepared title and abstract text produced in Section 4.2.3, was used as the input to a pretrained LLM accessed through an API. NER and RE were performed together within a single API request per abstract, with the model instructed to return entities and relationships following the BioRED extraction schema. The raw response was parsed into structured entity and relation predictions and stored for subsequent evaluation and processing.

4.4.1 LLM Configuration

All extraction was performed using a single model, OpenAI's `gpt-5.6-luna`, accessed via the OpenAI Python Software Development Kit (SDK)'s Responses API (`client.responses.create`). This pretrained model was used, without any parameter fine-tuning, across every EPI configuration, ensuring that any performance difference observed across EPIs resulted from changes to the pipeline's prompt and extraction setup rather than a change in the model's configuration. Higher-cost models, geared more towards complex reasoning over budget-consciousness, were considered for this project, such as `gpt-5.6-sol` or `gpt-5.6-sol`, however, they were deemed infeasible for this project given the number of API calls required across iterative EPI development and the full extraction runs on all four splits: the BioRED training, development and test splits, and the full CBC. No additional generation parameters were set for temperature or generation seed as `gpt-5.6-luna` did not allow either as a tuneable parameter to any of their reasoning models.

`gpt-5.6-luna`'s reasoning effort parameter (`reasoning` $\in$ {`low`, `high`, `medium`}), controlling the magnitude of internal reasoning before a response output, was also left at its default setting throughout this project, rather than being specifically tuned. Unlike model temperature, this was not a constraint of the model, but a deliberate choice made without deeper investigation into its effect on extraction quality and reasoning effort was not treated as a variable between EPIs. Increasing reasoning effort typically increases both time and cost demand per request, and investigating its effect on extraction precision, recall, and consistency was concluded to be out of the scope given the time and budget constraints discussed in Section 4.8. This parameter is one that a larger-budget project with longer timeframe allowance could reasonably investigate.

However, the absence of a temperature parameter and omission of fine-tuned reasoning parameter has a direct methodological consequence. Without these parameters, the model's output for a given prompt and abstract is not guaranteed to be identical across repeated calls, even when all else remains constant. This results in two distinct sub-consequences: firstly, as the reproducibility of an EPI's outputs across multiple identical calls is not guaranteed, any EPI that reuses its previous EPI's prompt reuses its stored outputs, rather than calling the model again. Secondly, an observed change in extraction performance between two EPIs using different prompt or extraction pipeline configurations cannot be attributed with complete certainty to the EPI refinement alone, as variation in performance may reflect natural variation in the model's generation, rather than just the effects of the update. This is partially mitigated by the fact that each EPI during the development stage was evaluated across the full BioRED training split (n = 400), rather than a small subsample of abstracts, which reduces the effects of variability of individual responses on performance metrics. Table 4.3 summarises the chosen model's extraction pipeline configuration.

Table 4.3: Extraction Model Configuration Summary

Parameter	Value
Model	`gpt-5.6-luna`
Provider / SDK	OpenAI Python SDK, Responses API
Fine-tuning	None
Temperature	Default (not explicitly set)
Reasoning	Default (not explicitly set)
Maximum output tokens	Default (not explicitly set)
Generation seed	Default (not explicitly set)
Request timeout	120s
Maximum automatic retries	2
Same model used across all EPIs	Yes

4.4.2 Prompt and Extraction Schema

EPI configurations were stored in externally from the project's main Python code as a set of dictionaries in JSON format including the EPI's name, prompt version — also stored externally in a prompt configuration JSON file: rather than embedding text directly into the extraction code, evaluation version — discussed in Section 4.4.3, and additional notes.

The prompt used by the project's EPIs was structured around several main components: a task definition instructing the model to extract entities and relationships exactly as they would be annotated under the official BioRED annotation guidelines, the BioRED guidelines themselves [18], an explicit list of the allowed entity and relationship types, evidence requirements, few-shot examples, and explicit output format instructions specifying the required JSON output. Each prompt version sequentially introduced one or more new components, initialised with the baseline prompt, including only explicit general instructions of the extraction task and the structured JSON output.

The permitted entity and relation types instructed in the prompt corresponded exactly to BioRED's own schema, given in Table 4.4.

The official BioRED annotation guidelines [18], retrieved from the corpus's public GitHub repository in PDF form, provides information of the corpus's own construction that are not relevant to instructing an LLM's extraction, such as the corpus's data sampling methodology, tools used in its ground truth construction, and its full reference list. Before incorporating the guidelines into the prompt, the document was therefore condensed to contain only the entity and relation annotation rules relevant to the extraction task itself. No semantic content within the filtered guidelines was altered — only its structure was changed, converting the document's bullet formatting into a consistently headed, nested bullet-point structure intended to be more easily parsed by the LLM when called. This condensed guideline version was stored in `.txt` format and embedded directly into the prompt as {`biored_ext_guidelines`}. The complete filtered BioRED extraction guidelines are provided in Appendix A.

Table 4.4: Permitted Entity and Relation Types

Entity types	Relation Types
ChemicalEntity	Association
DiseaseOrPhenotypicFeature	Bind
GeneOrGeneProduct	Comparison
SequenceVariant	Conversion
OrganismTaxon	Cotreatment
CellLine	Drug_Interaction
	Positive_Correlation
	Negative_Correlation

The evidence requirements instructed the model to base extractions only on information explicitly supported by the given abstract to avoid inferring entities or relationships from background biomedical knowledge rather than the abstract text itself, to avoid treating multiple entity mentions in the same sentence or context as automatically associated, and to avoid inferring a more specific relation type such as `Positive_Correlation` or `Negative_Correlation` than what the text explicitly included. The model was also instructed to exclude a relationship where it was uncertain if it satisfied these criteria. For each entity, the model was required to return its text and entity type, and for each relationship triple, two entities and a relation type.

Few-shot examples constructed directly from BioRED's own training split ground truth annotations, with each example including an abstract's text and its correct entity and relationship annotations in the same JSON structure the model was instructed to output. Three constant few-shot examples were included in each prompt by default. Once generated in its first inclusion in a prompt, the few shot block was exported and stored for subsequent inclusion, sharing the same configuration, rather than being resampled on every execution.

4.4.3 Structured Output and Parsing

The LLM was instructed through the prompt to return its extractions as a single JSON object containing two top-level keys: `entities` and `relationships`, each a list of extractions. No API-level constraint existed to enforce this structure, how closely the outputs adhered to the specified JSON output depended entirely on how well the model followed the prompt's explicit instructions.

Each abstract's raw response text was parsed as JSON in Python. Where a response failed to parse from JSON-structured string to valid JSON, this was recorded as a parse failure and the abstract was treated as having produced no entities and no relationships, rather than causing the run to fail. No further validation, such as checking that returned entity or relation types belonged to the allowed schema, was conducted beyond this JSON-validity check. Successfully parsed entities and relationships were converted into tuples following the form `(PMID, entity text, type)` for entities and `(PMID, entity 1, relation, entity 2)` for relationships. Entity text was stripped of any excess whitespace and stored in lowercase. These predictions sets were then converted into the same tabular structure used for BiORED's own ground-truth entities and relations, described in Section 4.2.3, allowing predicted and ground-truth extractions to be compared directly during evaluation.

Finally, after each EPI's final extraction run on the entire BioRED train set, the run was exported as its own run in full, storing the EPI's identifier (e.g.: `epi_001`), the dataset split it was evaluated on, the prompt and evaluation version used, the number of abstracts processed, the count of failed JSON parses, the raw prompt template, each abstract's raw total model output, the parsed entity and relation predictions, and the evaluation metrics and errors resulting from that run. The raw and parsed extractions were stored as, as previously mentioned, when a new EPI reused the previous EPI's prompt version and only updated the evaluation approach, the previous EPI's outputs were used — storing outputs allowed for the importing of previous EPI outputs when necessary. How these logged runs were compared and used to select a final EPI is described in Section 4.5. The complete text of each prompt version is provided in Appendix B.

4.5 Pipeline Development

Development of the extraction pipeline was organised as a sequence of Extraction Pipeline Iterations EPIs. Each EPI represented a pre-configured combination of a prompt version and evaluation version and was assigned a unique identifier (e.g., `epi_001`) used to track it during development. Changes between EPIs were introduced incrementally, with each new EPI modifying one or a few components of the previous iteration, so that the effect of an individual methodological change could be isolated, rather than observing the effects of several significant changes. As described in Section 4.1, all EPI development and refinement was conducted exclusively against the BioRED training split, with the BioRED development split used for the comparison of candidate EPIs, playing no role in extraction pipeline refinement. Each EPI's full configuration, prompt template, and extraction outputs were automatically logged after extraction, allowing outputs to be reused directly by a subsequent EPI where prompt version remained unchanged between them.

4.5.1 Extraction Pipeline Iterations

Table 4.5 summarises the configuration of all EPIs produced during development, including the main methodological change in addition to the previous iteration.

Table 4.5: Extraction Pipeline Iteration summary

EPI	Prompt	Evaluation	Main addition/purpose
`epi_001`	`v1`	`v1` (strict matching)	Baseline configuration
`epi_002`	`v1`	`v2` (relaxed matching)	Relaxed matching to baseline prompt
`epi_003`	`v2`	`v2`	Filtered BioRED guidelines and allowed entity and relation types
`epi_004`	`v2`	`v3` (cosine matching)	Cosine matching
`epi_005`	`v3`	`v3` (cosine matching)	Few-shot examples
`epi_006`	`v3`	`v4` (relation order invariance)	Relation order invariant evaluation
`epi_007`	`v4`	`v4`	Targeted suppression of background biomedical knowledge
`epi_008`	`v5`	`v4`	Explicit instructions preventing inference of an unsupported extraction
`epi_009`	`v6`	`v4`	Explicit precision-over-recall instructions
`epi_010`	`v7`	`v4`	Precision-focussed conservative extraction instructions
`epi_011`	`v8`	`v4`	Targeted suppression of excessive relation over-extraction

Together, this sequence of pipeline iterations reflects several broad stages of development, rather than a linear progression of independent changes: an initial baseline extraction stage (`epi_001-002`), alignment of the prompt with BioRED's own entity and relation schema (`epi_003-004`), introduction of few-shot prompting (`epi_005`), concurrent evaluation logic progression (`epi_001–006`), tightening of evidence requirements (`epi_007–008`), and a final stage of prompt refinement targeting over-extraction (`epi_009–011`). It should be noted that, for each EPI where evaluation methodology was updated, the previous EPI's prompt outputs were used. This allowed direct analysis of the evaluation refinement's effects on extraction performance metrics.

As mentioned above, evaluation methodology applied to each EPI's outputs also evolve over the course of development, concurrently with prompt refinement. As summarised in Table 4.5, the baseline EPI was assessed using strict matching, where only exact string matching between an extraction and a ground-truth counted as a success. Manual inspection of errors — automatically recorded in each EPI's stored log, including FPs and FNs — indicated that numerous extractions represented the correct biomedical concept, yet differed from the ground-truth annotation in span boundary or string representation, meaning strict matching classified some semantically correct predictions as errors. This motivated the introduction of relaxed matching, which allowed correct-match classification where the string representation of the extracted entity differed slightly from that of its corresponding ground truth. However, this implementation was found to be insufficient where an extraction and its ground truth were semantically equivalent but expressed using disparate string representations, exceeding solely a mismatch between entity name boundaries. As a result, subsequent cosine-similarity-based matching was introduced, which compared the vector embeddings of extracted and ground truth entity mentions using a fixed threshold. The exact configuration and implementation of strict, relaxed, cosine, and relation order invariance evaluation approaches are described in Section **??**.

When necessary, to understand the nature of extraction errors, false positive (FP) and false negative (FN) relation errors produced by the current EPI were sampled and analysed. This analysis was restricted to relations, rather than entities, as relation extraction exhibited significantly weaker performance throughout development and was therefore the main focus of this investigation. For the selected EPI, up to 50 false positive FP and 50 false negative FN errors were randomly sampled, using a fixed random seed for reproducibility, and assigned one of a set of pre-defined error categories.

Given the volume of errors sampled, manual assignment of each error to a pre-defined error category was not feasible. Instead, categorisation was performed automatically using the same LLM used for the extraction process. For each sampled error, the model was provided with the incorrect relation, the error type (FN or FP), the source abstract, and the corresponding BioRED ground truth relations for that abstract. The model was then instructed to assign the error to exactly one of seven predefined error categories (Table 4.6), each including a description of the conditions of when the error category is true. The complete prompt supplied to the LLM during error analysis is provided in Appendix C.1.

Table 4.6: Error category summary

Category	Description
Wrong relation type	”The correct entity pair is identified, but the predicted relation type differs from the BioRED ground truth.”
Duplicate/coreference	”The error results from duplicate entity mentions, abbreviations, aliases, or coreferential mentions of the same underlying entity.”
Unsupported relation	”The predicted relation is not supported by the abstract or BioRED annotation policy.”
Entity boundary/synonyms mismatch	”The prediction and ground truth refer to the same underlying entity, but differ in span, wording, abbreviation, or synonym representation.”
Missing relation	”A BioRED ground-truth relation is supported by the abstract but was not extracted by the system.”
Evaluation artefact	”The prediction is substantively compatible with the ground truth, but is counted as an error because of the evaluation or matching procedure.”
Other/unclear	”The primary cause cannot be confidently assigned to any other category.”

These category distributions were used diagnostically, precision, recall, and F1 to establish whether an EPI’s performance reflected true extraction failure, evaluation artefacts, or specific prompt configuration issues. Analysis showed that unsupported relations was a significant false-positive category through much of development, directly motivating the evidence requirement prompt refinements in `epi_007–epi_011`. A significant proportion of false negatives were also categorised as evaluation artefacts, supporting the introduction of cosine-based matching. Figure 4.2 shows a representative error distribution figure from one stage of this analysis (epi_009) — the remaining error distribution figures are shown in Appendix C.2.

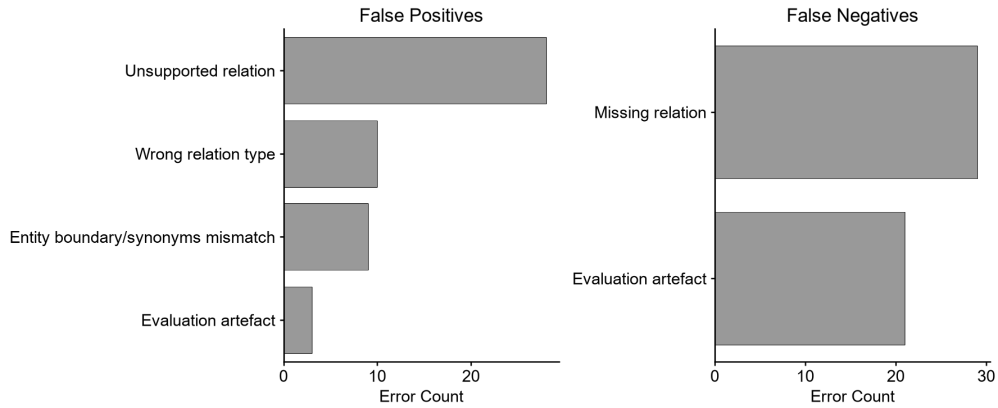

Figure 4.2: `epi_003` relation error distribution.

4.5.2 Development and Model Selection

Iterative EPI development against the BioRED training split contiued until further prompt or evaluation refinements produced only marginal changes in extraction performance metrics. At this point, all candidate EPI configurations were frozen and no further prompt or schema adjustment was made.

Figure 5.1 in Section **[NEED REFERENCE]** shows the entity and relation precision, recall, and F1 across all EPIs measured on the BioRED development split, after being refined on the training split, allowing performance to be tracked across sequential extraction pipeline refinements. On the basis of primarily F1, and secondarily precision and recall, `epi_007`, `epi_008`, and `epi_009` were selected for further comparison.

Figure 5.2 (Section **[NEED REFERENCE]**) compares these three selected EPIs directly on relation precision, recall, and F1, again, evaluated on the unseen BioRED development split. Final selection between the three candidates was mainly based on relation F1, as relation extraction was the main target of refinement at this stage of development. `epi_009` was selected as it exhibited the highest F1 and precision, while narrowly underperforming compared to `epi_007` on recall. It should be noted that while `epi_009` marginally outperformed its competitors on balance, the differences in extraction performance for all three performance metrics were near negligible.

No further prompt, schema, or evaluation changes were made to the selected EPI following this comparison. The resulting configuration was therefore frozen and used without modification for the final BioRED benchmark evaluation and extraction of the CBC.

4.5.3 Final Pipeline

The final extraction pipeline configuration selected through the process above is summarised in Table 4.7. This configuration therefore represents the final extraction system evaluated in all subsequent experiments reported in this thesis.

Table 4.7: Final Extraction Pipeline Configuration

Component	Final configuration
EPI	`epi_009`
Prompt version	`v6` (Suppression of background knowledge, unsupported extraction inference suppression)
Evaluation version	`v4` (Cosine-based, relation order invariant)
LLM	`gpt-5.6-luna`
Few-shot examples	3 fixed BioRED training examples
Entity schema	BioRED six entity types (Table 4.4)
Relation schema	BioRED eight relation types (Table 4.4)
Output format	JSON object

4.6 Extraction Evaluation

The final EPI selected in Section 4.5 was evaluated using a single evaluation pipeline, applied without modification to both the held-out BioRED test split and the CBC. This pipeline used the same prompt, extraction logic, and evaluation configuration for both datasets, differing only in which abstracts were passed to the pipeline and which PMIDs were included when evaluating extraction performance. In both cases, extraction was first performed over every abstract, then extractions were restricted to a defined set of PMIDs before being compared against ground truth. For the BioRED test evaluation, this evaluation subset was identical to the full test split as all abstracts in the BioRED test set had existing ground truth annotations. For the CBC, extraction was instead performed over the full 282-abstract corpus while the evaluation subset was restricted to the 50 abstracts that had been randomly sampled from the full CBC dataset and manually annotated for both NER and RE ground truths. Predictions from the remaining, unannotated CBC abstracts were excluded from evaluation. Consistent with the development process, entity and relation performance were evaluated separately, with precision, recall, and F1 as the primary performance metrics:

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where a TP is an extracted entity or relation matched to a ground truth annotation under the matching approach mentioned previously, an FP is a predicted item that did not match with any of the abstract's ground truths, and an FN is a ground truth annotation that was not extracted by the model. For both the BioRED test split and CBC dataset, point estimates were calculated, including a 95% confidence interval from bootstrap sampling of document level TP/FP/FN counts with a resampling size of 2,000. As the same matching procedure, embedding model, cosine similarity threshold, and bootstrap sampling approach was used for both datasets, entity and relation results are directly comparable between them.

4.6.1 Entity Evaluation

Within the evaluation dataset described above, predicted entities were compared against ground truth entity annotations only within the same document by PMID. Each entity was represented as a `(PMID, text span, entity type)` tuple, with text stripped of whitespace and converted to lowercase before comparison. A predicted entity type could only match a ground truth annotation only if both the PMID and entity type matched exactly.

For entities that satisfied these criteria, text span similarity was assessed using vector embedding based evaluation. Both the predicted and all candidate match ground truth text spans were encoded using the `pritamdeka/S-PubMedBert-MS-MARCO` sentence-embedding model, a PubMedBERT-based model tuned for biomedical sentence similarity. Matches were then validated using cosine similarity:

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{(\mathbf{u} \cdot \mathbf{v})}{(\|\mathbf{u}\| \|\mathbf{v}\|)}.$$

A predicted entity was considered a match for a candidate ground-truth entity if this similarity was greater or equal to the predefined threshold of 0.85.

One-to-one matching was followed for each abstract–title pair where for each predicted entity, every remaining candidate unmatched ground-truth entity was considered, and the prediction was matched to the ground truth annotation that gave the greatest cosine similarity greater than or equal to the threshold. Subsequently, this ground truth entry was removed from matching future predictions. Matched predictions were counted as TPs, predicted entities that did not match a ground truth as FPs, and ground truth entities that were unmatched after the abstract's matching step were counted as FNs. These counts formed the basis of precision, recall, and F1.

4.6.2 Relation Evaluation

Each predicted and ground truth relation was represented as a `PMID, entity_1, relation, entity_2` tuple, again compared to extractions of the same paper by PMID and within the evaluation subset. Relation correctness depended on both the exact relation label and on both entity spans. For example, correctly identifying two correct entity spans, but with an incorrect relation, did not count as a match. As the extracted relation label had to be exactly the same string representation as its corresponding ground truth, no similarity-based evaluation was applied to the relation label itself. Consistent with the entity evaluation, both predicted relation entities were matched against candidate ground truths using the same PubMedBERT vector embeddings and cosine threshold of 0.85 defined in Section 4.6.1.

As mentioned in Section 4.5.1 and highlighted in Table 4.5, final relation matching was order-invariant and applied for all relation types, where a predicted relation and ground truth pair was recorded as a match if either the predicted relation matched its corresponding ground truth entities in either the same or reversed order. Formally, for a predicted relation (e_1, r, e_2) and ground truth relation (g_1, r, g_2), a match was recorded when either $e_1 = g_1$ and $e_2 = g_2$ or $e_1 = g_2$ and $e_2 = g_1$.

Before relation order invariance:

$$\text{(warfarin, Positive\_Correlation, ICH)} \neq \text{(ICH, Positive\_Correlation, warfarin)}$$

After relation order invariance:

$$\text{(warfarin, Positive\_Correlation, ICH)} \equiv \text{(ICH, Positive\_Correlation, warfarin)}$$

Before relation order invariance was implemented, relations were manually reviewed for directionality. None of the eight relation labels were considered unidirectional, thus relation matching was treated as order-invariant for all relation types.

Consistent with entity matching, one-to-one matching was enforced per document, with each predicted relation being matched with the highest-scoring unmatched candidate ground truth relation meeting the criteria described in Section 4.6.1. Again, matched predictions were counted as TPs, unmatched predictions as FPs, and unmatched ground truth relations as FNs, from which precision, recall, and F1 were also computed. As a relation match requires both entity spans and the relation label to be extracted correctly, relation extraction was overall a more demanding matching criteria than entity extraction. This may partly be why relation extraction performance was consistently poorer than that of entity extraction.

4.6.3 CBC Manual Annotation

Given the complexity of biomedical abstracts and the authors lack of biomedical expertise, and time constraints, it was decided that fully manual annotation of the CBC abstracts downsample was infeasible.

During construction, the original BioRED dataset used a semi-automated approach, whereby entity annotations were initially generated using PubTator Central, a biomedical RE service [31], and subsequently reviewed and corrected by expert human annotators. Consistent with the original BioRED ground truth annotation construction, the same approach was adopted for the CBC, using automated entity extraction as an initial annotation layer followed by manual verification and correction.

Automatic pre-annotation of the CBC's 50 downsampled abstracts initially considered using PubTator Central, and subsequently PubTator3, NCBI's enhanced successor to their PubTator Central service [32], given its underlying BioREx extraction model shares its training foundations with BioRED's own entity and relation schema, making it a suitable candidate to uphold consistency between BioRED-based development and CBC evaluation ground truth schema. However, as PubTator3 exists as a biomedical entity extraction service that returns entities from a comprehensive database, rather than an on-demand API-accessible extraction pipeline, it lagged behind the most recent biomedical literature. This was demonstrated during initial attempted entity pre-annotation as annotations were disproportionately concentrated in the earlier years of the CBC's 2024-2026 publication range, rather than evenly distributed across it. To overcome this, NCBI's All-in-One scheme-based Biomedical Named Entity Recognition (AIONER) model [33], the underlying model used for PubTator3 entity recognition, was used.

The 50 downsampled abstracts were first exported into a PubTator file, allowing for direct input to AIONER. The AIONER model was then used to perform automatic entity recognition over these abstracts, producing candidate entity pre-annotations together with their span boundaries and entity types. AIONER's native entity labels did not exactly match the six BioRED entity types, for example, `Chemical` rather than BioRED's `ChemicalEntity`, and `Variant` rather than BioRED's `SequenceVariant`, and were therefore mapped onto the corresponding BioRED entity types during manual review.

Additional manual checks and inspection was conducted: for each abstract, the original abstract text was read directly against the AIONER's pre-annotations to validate or map entity type, identify missing entities, correct incorrect span boundaries — such as `adeno` instead of the correct `adeno-associated virus`, and remove incorrectly extracted pre-annotations. To reduce human error arising from manually correcting entity types, Microsoft Excel data validation was used to enforce fixed selection, rather than free-text entry. This process produced the finalised CBC NER ground truth.

Relation annotation was then conducted using this finalised entity set. As with entity annotation, Excel data validation was used to reduce human error, in this case using an Excel lookup function referencing the finalised list of NER spans, ensuring that every entity referenced in a relation corresponded exactly to an already-validated entity text span, rather than an incorrectly manually typed version. As with manual CBC NER annotation construction, the original abstract text was inspected in order to identify any possible entity relations explicitly supported by the abstract text. This produced the finalised CBC RE ground truth.

Both resulting datasets were structured in the same tabular format as the corresponding BioRED data splits, given in Tables 4.8 and 4.9 below. This meant that existing extraction parsing and evaluation logic described previously required no modification for CBC NER and RE extraction and subsequent evaluation.

Table 4.8: Extract of the manually annotated CBC named entity recognition ground truth format.

PMID	Text Span	Entity Type
40655353	Acinetobacter baumannii	OrganismTaxon
40655353	infections	DiseaseOrPhenotypicFeature
40655353	macrolides	ChemicalEntity
...	...	...

Table 4.9: Extract of the manually annotated CBC relation extraction ground truth format.

PMID	Entity 1	Type 1	Relation	Entity 2	Type 2
40655353	Oxa	GeneOr GeneProduct	Negative_ Correlation	carbapenem	Chemical Entity
40655353	Oxa	GeneOr GeneProduct	Negative_ Correlation	cephalosporin	Chemical Entity
40655353	Oxa	GeneOr GeneProduct	Negative_ Correlation	penem	Chemical Entity
...	...	...	...	...	...

4.6.4 Generalisation Evaluation

As described above, for the CBC, evaluation was restricted to predictions from the 50 abstracts comprising the manually annotated CBC ground truth subset, using the procedures defined in Sections 4.6.1 and 4.6.2, allowing entity and relation precision, recall, and F1 to be evaluated on CBC extractions and compared directly against the corresponding BioRed test set results.

Lower measured performance o the CBC relative to the BioRED test split is interpreted as possible evidence of reduced generalisation to contemporary, independently and automatically sampled biomedical literature. This shift should not be attributed solely to a temporal shift in the data, as the two corpora also differ in construction, sampling, and, most significantly, annotation procedure. Further discussion of possible performance difference is given in Chapters 5 and 6.

Unlike the relation-focussed analysis during development, the LLM-based CBC error analysis additionally extended to entity FPs and FNs, sampled and categorised using the same seven error categories (Table 4.6), allowing entity and relation error distributions to be analysed side by side for extraction evaluation. As during development, up to 50 FP and 50 FN errors were randomly sampled using a fixed random seed, this time for both entities and relations. Each sampled error was categorised automatically using the same LLM-based categorisation process described in Section 4.5.1, given the incorrect extraction — comprising text span and entity type, and its corresponding paper's abstract text, title text, and ground truth annotations to provide context. The resulting category distributions were visualised, providing insight for assessing whether any performance degradation of CBC extractions reflected genuine extraction failure or evaluation artefacts.

Table 4.10 summarises the evaluation procedure for both the CBC and BioRED test split, and the final generalisation approach.

Table 4.10: Final Evaluation Summary

Evaluation Type	Matching requirements	Metrics
Entity	Same document exact entity type match, entity span cosine similarity ≥ 0.85, one-to-one matching	Precision, Recall, F1 (95% bootstrap CI)
Precision	Same document exact relation type match, entity span cosine similarity ≥ 0.85, order invariance, one-to-one matching	Precision, Recall, F1 (95% bootstrap CI)
Generalisation	Identical evaluation procedure applied to CBC extractions, restricted to the manually annotated CBC subset (n = 50 of 282)	Precision, Recall, F1

4.7 Contemporary Knowledge Graph Construction

4.7.1 Named Entity Normalisation

Following extraction, entity mentions produced by the final EPI (Section 4.5.3) from the CBC were passed through Named Entity Normalisation NEN, converting each free-text mention and its entity type into a standardised concept identifier drawn from the biomedical terminology resources used by the BioRED dataset, prior to ontology-based validation (Section 4.7.2). The extraction stage represents each entity only as a text span and an entity type, with no associated identifier. NEN normalises the entity mentions so that the difference text representations of the same biomedical concept, such as synonyms, abbreviations, or alternate terminology, are consolidated into a single identifier, and therefore a single final graph node in the resulting Knowledge Graph KG.

NEN was applied only to extracted entities that were included in at least one predicted relation, rather than to every entity extracted by the pipeline. Entities that were extracted but never appeared in an extracted relation triple would only form isolated orphan nodes — nodes that are not linked by edges to any other node. Therefore, normalisation of these entities was decided unnecessary use of any terminology resource APIs.

For each target abstract, every unique `(PMID, entity text span)` pair appearing at least once in any relation was identified from the extraction pipeline's CBC stored predictions. Each mention was then matched to its corresponding entity prediction to extract its entity type, which determined the appropriate terminology resource for entity normalisation. The entire set of `(text span, entity type)` pairs was then stripped of any duplicates across abstracts, so that extractions of the same entity–type pair across abstracts was normalised once, instead of multiple times.

The Unified Medical Language System UMLS Metathesaurus was considered as the primary terminology resource. However, the decision was made to instead access several type-appropriate biomedical resources, as the UMLS Metathesaurus required a licenced account for full access **[MORE DEPTH]**. The terminology resources used are summarised in Table **[NEED REFERENCE]**. **[NEED REFERENCE]**

Table 4.11: Entity Type–Normalisation Resource Mapping

BioRED entity type	Normalisation resource	Access method
`GeneOrGeneEntity`	NCBI Gene	NCBI E-utilities API (`db=gene`)
`ChemicalEntity`	MeSH	NCBI E-utilities API (`db=mesh`)
`DiseaseOr PhenotypicFeature`	MeSH	NCBI E-utilities API (`db=mesh`)
`OrganismTaxon`	NCBI Taxonomy	NCBI E-utilities API (`db=taxonomy`)
`CellLine`	Cellosaurus	NCBI Cellosaurus REST API
`SequenceVariant`	dbSNP / NCBI Variation Services	NCBI E-utilities API/Variation Services

4.7.2 Ontology-Based Validation

4.7.3 Graph Construction

4.8 Cost Analysis

Chapter 5

Results

Table 5.1: Entity Extraction Performance

EPI	Reported matching approach	Precision	Recall	F1
epi_001	Strict	0	0	0
epi_002	Relaxed	0	0	0
epi_003	Relaxed	0.865	0.774	0.817
epi_004	Cosine	0.894	0.800	0.845
epi_005	Cosine	0.930	0.736	0.821
epi_006	Cosine + relation order invariance	0.930	0.736	0.821
epi_007	Cosine + relation order invariance	0.918	0.762	0.833
epi_008	Cosine + relation order invariance	0.901	0.765	0.828
epi_009	Cosine + relation order invariance	0.935	0.761	0.839
epi_010	Cosine + relation order invariance	0.941	0.754	0.837
epi_011	Cosine + relation order invariance	0.932	0.738	0.824

Table 5.2: Relation Extraction Performance

EPI	Reported matching approach	Precision	Recall	F1
epi_001	Strict	0	0	0
epi_002	Relaxed	0	0	0
epi_003	Relaxed	0.192	0.189	0.190
epi_004	Cosine	0.423	0.416	0.420
epi_005	Cosine	0.392	0.419	0.405
epi_006	Cosine + relation order invariance	0.478	0.511	0.494
epi_007	Cosine + relation order invariance	0.495	0.524	0.509
epi_008	Cosine + relation order invariance	0.487	0.516	0.501
epi_009	Cosine + relation order invariance	0.502	0.516	0.509
epi_010	Cosine + relation order invariance	0.501	0.488	0.494
epi_011	Cosine + relation order invariance	0.501	0.456	0.478

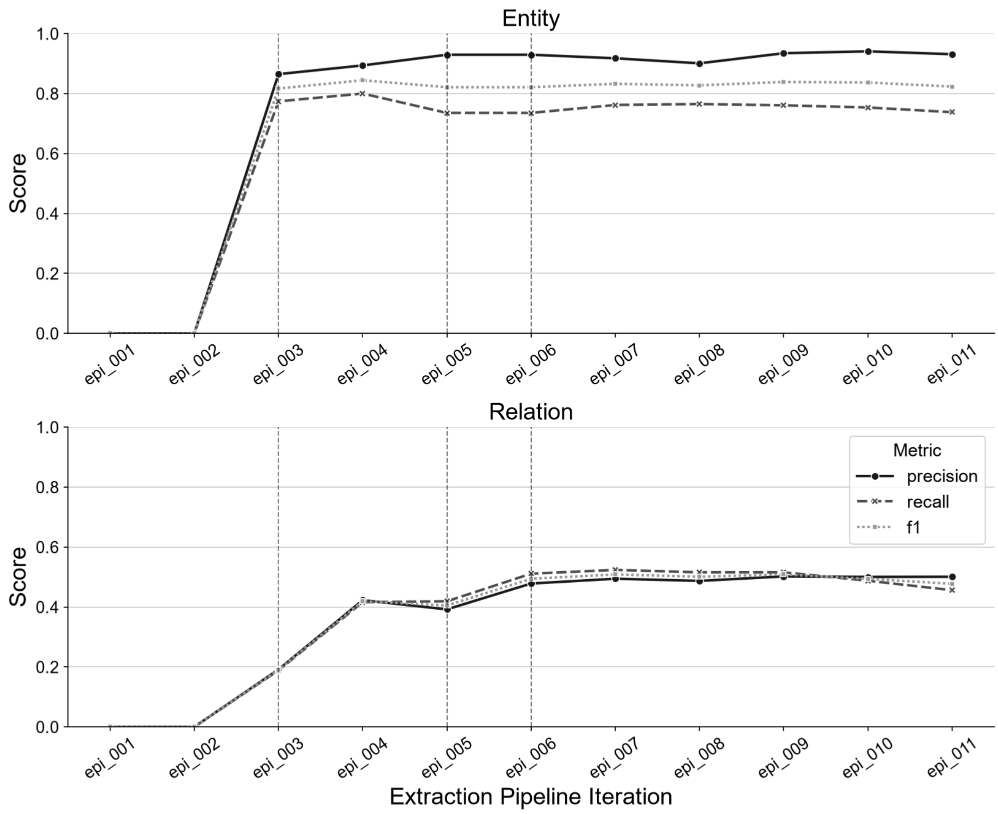

Figure 5.1: EPI Performance

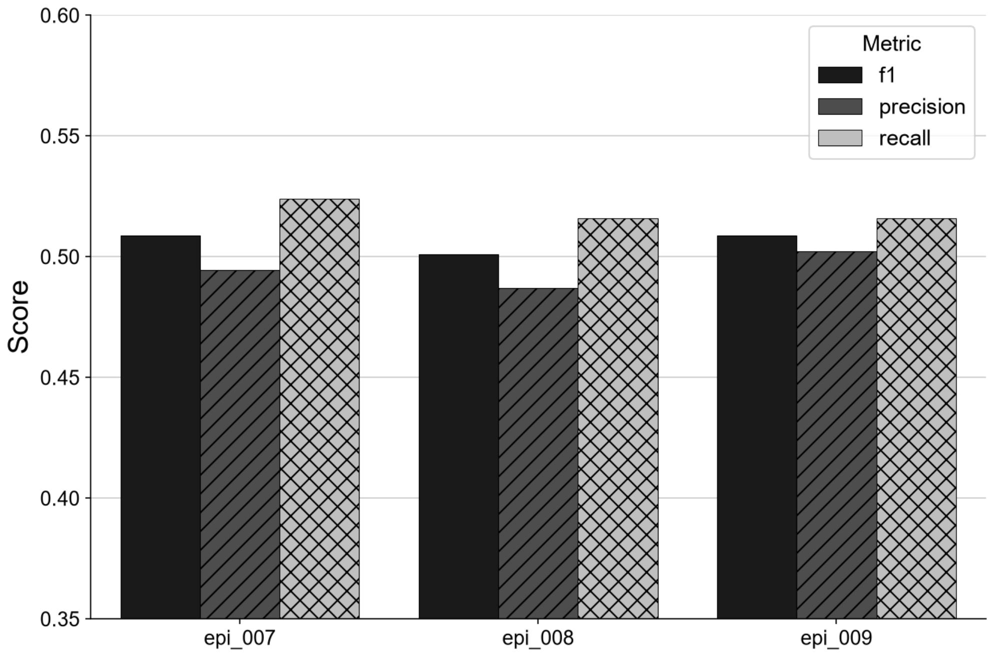

Figure 5.2: Selected three best-performing EPIs

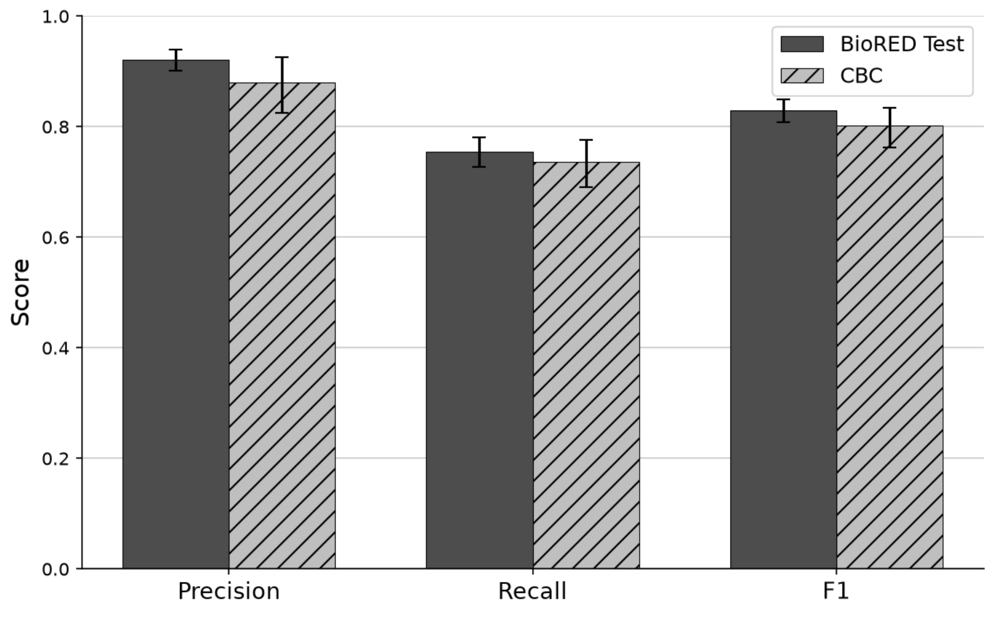

Figure 5.3: Entity Generalisation: BioRED vs. CBC

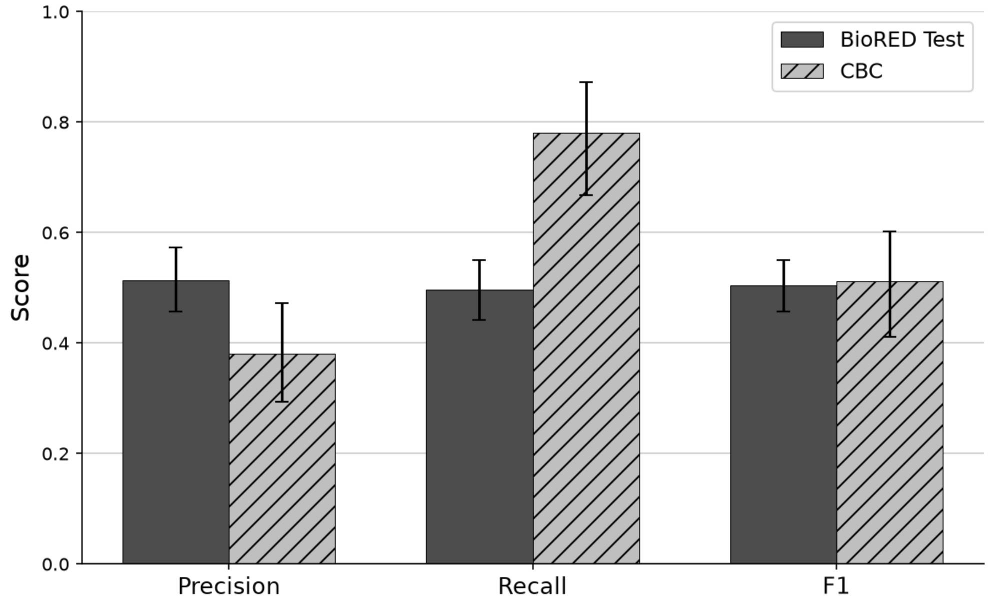

Figure 5.4: Relation Generalisation: BioRED vs. CBC

Chapter 6

Discussion

Chapter 7

Conclusion

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Appendix A

Filtered BioRED Annotation Guidelines Used For Prompting

```
## Guideline of the entities

### General rules
- Annotate all the spans of all the six concept types.
- The full text can be accessed to clarify the concept spans and
   identifiers.
- The abbreviation and its long form should be annotated
   separately if possible. prostaglandin E2 (PGE2) in the text, "
   prostaglandin E2" and "PGE2" should be both annotated to
   chemicals with the same identifier (D015232).
- Annotate both the full name and abbreviation in one entity, if
   the boundary of the entity covers both of them. For instance,
   annotate "Deoxyguanosine kinase (dGK) deficiency" entirely to
   a disease (PMID:19394258).
- Annotate the composite entity spans (e.g., "MMP-1, -2") entirely
   . The identifiers of the multiple concepts should
be entered separated by "," with no space (e.g., "4312,4313").
- Annotate any concept identifier span (e.g., "OMIM 307800" and "
   rs729302").
- Annotate the concept spans with misspelling ("HLRRC" in PMID
   :18366737).

### Gene
- The gene spans should be linked to NCBI Gene ID (https://www.
   ncbi.nlm.nih.gov/gene/). Specify the corresponding species and
    the identifiers to the gene spans.
- Annotate the gene families including paralogs/orthologs (e.g.,
   bone morphogenetic protein(bmp)), but DO NOT annotate the
   functional based families (e.g., tumor suppressor gene,
   protein kinase).
- Following the previous rule, if at least one of the gene family
   members mentioned in the text, the identifiers of those
   members are assigned to the family name (e.g., in "Cytochrome
   P-450 genes (CYP1A1, CYP2A6, CYP2D6, and CYP2E1)", Cytochrome
   P-450 genes should be assigned with NCBI Gene:
```

```
    1543,1548,1565,1571). If no family member is in the text, the
    first gene reached in the gene family should be assigned (e.g.,
     matrix metalloproteinases to NCBI Gene:4312(MMP1)).
- In experimental studies, annotate the gene identifier
    corresponding to the organism/species that is the source of
    the
gene (for example mouse gene transfected into human cells is
    annotated to record for mouse gene).
- DO NOT annotate the genes, if they are named for cells and
    pathways.

### Disease
- The diseases refer to MEDIC which is a combination of MESH (
    https://meshb.nlm.nih.gov/search) and OMIM (https://omim.org/).
     We annotate the MESH identifier, if a disease has both the
    MESH and OMIM identifiers.
- If no specific disease concept can be reached in MEDIC by the
    annotated span, we annotate the closest hypernym concept that
    logically describes the disease span(e.g., X-linked retinitis
    pigmentosa to D012174 (retinitis pigmentosa) in PMID:17935240).
     Otherwise, "-" is assigned to the mismatched diseases.
- Annotate the diseased-related phenotypes but not other (e.g.,
    short ear in PMID:17345627).
- DO NOT annotate the high-level diseases (e.g., "genetic disorder
    ").
Excludes:
- Cis-acting disease (PMID:12442272)
- genetic disorders (PMID:18046082)
- familial disorder (PMID:20806042)
- Annotate "drug addiction" to "disease".
- Annotate the symptoms to diseases.
- DO NOT annotate overdose.
- DO NOT annotate the mental status and phenotypes to diseases
    unless the term is exactly recorded in MEDIC. e.g., decline in
     sympathetic tone (PMID:19067809)
- DO NOT annotate references to biological processes such as "
    tumorigenesis" or "cancerogenesis".
- Annotate minimum necessary text spans for a disease. For example
    , select "hypertension" instead of "sustained hypertension.",
    unless the span covers the prefix or the suffix can exactly
    match a specific concept identifier (e.g., "chronic bronchitis"
     to D029481).

### Chemical
- Chemical spans should be linked to MESH (https://meshb.nlm.nih.
    gov/search). "-" is assigned to the mismatched chemicals.
```

```
- Annotate the chemical abbreviation with the dose. For example " In addition, GABA content of mice hippocampus treated with GFC75 plus P400 showed an increase of 46.90% when compared with seized mice.", 75 and 400 suffixes are the doses. In this case, we annotate GFC75 and P400 as chemical spans. (PMID:24911645).
- Annotate the chemical resource for extracts, e.g., "ethanolic extract of Daucus carota seeds (DCE)" in PMID:16755009; "grape seed proanthocyanidin extract" in PMID:11334364.
- DO NOT annotate antibodies to chemicals. (PMID:17595233) (PMID:20648600)
- DO NOT annotate saline, placebo, water, and juice.
- DO NOT annotate residue, free radicals.
- DO NOT annotate staining chemical (e.g., hematoxylin in PMID:24442316)
- DO NOT annotate vaccines.

### Species
- The species spans should be linked to NCBI Taxonomy ID (https://www.ncbi.nlm.nih.gov/taxonomy/)
- DO NOT annotate beyond species rank (e.g., annotate homo sapiens but do not annotate mammalian which is a taxonomy class)

### Variant
- The annotation of the variants follows tmVar annotation guidelines, the variants should be linked to dbSNP (https://www.ncbi.nlm.nih.gov/snp) accession number (RS#) when possible. Otherwise, we annotate the variant components (e.g., wild type, location, and mutant) following the tmVar component format (e.g., V600E to "p|SUB|V|600|E").
- DO NOT annotate residue of the gene modification.
- For those variants that cannot be normalized: In our observations on the variant recognition/normalization dataset, around 50-60% of the variants cannot be normalized to the specific concept identifiers in dbSNP. For those, we enumerate the components of the variants.

### Normalized components of the curated variant types:

#### Variant on DNA Sequence
Example(s) and normalized component or identifier(s):
- c.1922G>A: c|SUB|G|1922|A
- C-to-G transition was identified at nucleotide 857: c|SUB|C|857|G

#### Variant on protein sequence
```

```
Example(s) and normalized component or identifier(s):
- R114H: p|SUB|R|114|H
- methionine to threonine substitution at codon 235: p|SUB|M|235|T

#### RS number
Example(s) and normalized component or identifier(s):
- rs763780: rs763780

#### Allele on DNA sequence
Example(s) and normalized component or identifier(s):
- -281G: c|Allele|G|-218

#### Allele on protein sequence
Example(s) and normalized component or identifier(s):
- L638: p|Allele|L|638
- stop codon at position 372: p|Allele|X|372

#### Nucleotide or acid change
Example(s) and normalized component or identifier(s):
- G > C: c|SUB|G||C
- valine for glutamate: p|SUB|V||E

## Guideline of the relation pairs

### General rules
- We only annotate the (1) Popular associations and the (2)
    Corresponding gene/species in Figure 1.
- Co-occurrence in the same sentence is not required.
- DO NOT annotate a confirmed irrelevance. (e.g., "Nonetheless,
    NBS1 gene heterozygosity is not a major risk
factor for lymphoid malignancies in childhood and adolescence" in
    PMID:16152606)
- Annotate the associations with explicit statements in the text.
- The full text is NOT allowed to access. DO NOT annotate if the
    statement in the abstract is not clear. (e.g.,
PMID:14722929)
- For Co-treatment, we annotate "Negative_Correlation" between the
     two chemicals and the disease, and a
"Cotreatment" relation between the two chemicals. For an example
    in PMID:16720068, "Possible neuroleptic
malignant syndrome related to concomitant treatment with
    paroxetine and alprazolam.".
- For Drug_Interaction, we annotate "Positive_Correlation" between
     the two chemicals and the disease, and a
"Drug_Interaction" relation between the two chemicals.
```

```
### Disease-chemical
- Annotate the association of chemicals and disease due to the "
    toxicity" and "drug-induced", but not annotate the
chemicals with the disease concept-"toxicity" directly.
    - PMID:19728177
    "RESULTS: Our patient was admitted to the MICU after being
        found unresponsive with presumed toxicity from
        acetaminophen which was ingested over a 2-day period. ...
        In patients with FHF and cerebral edema from acetaminophen
        overdose, prolonged therapeutic hypothermia could
        potentially be used as a life saving therapy and a bridge
        to hepatic and neurological recovery." We annotate the
        associations between acetaminophen and FHF/cerebral edema
        due to the toxicity, but we do not annotate acetaminophen
        and toxicity.
- Annotate the relation between the target disease and the measure
     level of the molecules/chemicals in blood test
with correlation, "Compared with control patients, CRF and ESRD
    patients had higher preoperative serum
creatinine levels, a greater percentage of patients with
    hepatorenal syndrome.", the pairs of "CRF, creatinine" and
"ESRD, creatinine" is annotated. (PMID:11773892)
- DO NOT annotate the association of disease to the staining
    chemicals. For instance, "The extent of neuronal injury
was determined by 2,3,5-triphenyltetrazolium staining." (PMID
    :1711760) and "Renal lesions were analyzed in
hematoxylin and eosin, periodic acid-Schiff, and Masson's
    trichrome stains. SRL-treated rats presented proteinuria
and NGAL (serum and urinary) as the best" (PMID:24971338)
- DO NOT annotate the association of any disease with the chemical
     concept "analgesia" in "patient-controlled
analgesia (PCA) pump". For instance, "patient-controlled analgesia
     (PCA) pump" (PMID:9672936)
- DO NOT annotate toxicity with its chemical. For instance, do not
     annotate "unresponsive with presumed toxicity
from acetaminophen" (PMID:19728177)
- Annotate the chemicals which prevents the chemical-induced
    disease (PMID:24971338+18503483)

### Disease-gene
- DO NOT annotate the prognostic factor of genes or its variants
    to the target disease.
- Annotate the negative correlation (e.g., knockdown gene to the
    target disease). For instance, annotates Gankyrin
and HCC in "Gankyrin expression in the tumor microenvironment is
    negatively correlated with progression-free
```

```
survival in patients undergoing sorafenib treatment for HCC." (
    PMID:28777492)
- DO NOT annotate the symptoms of the target disease to the
    correlated variants or genes, unless the clear statement
between the symptom and variant is provided.

### Disease-variant & Disease-gene
- Annotate the corresponding diseases with the variant when it is
    observed in patients.
- Annotate the diseases with the associated genes.
- Annotate the corresponding gene of the variant with the variant-
    associated disease.
- Annotate the association between the variants (and corresponding
     gene) and the diseases belonging to the target
disease. For instance, annotate the association between "autosomal
     recessive disease" and "272gly----stop"
(PMID:1671881).
- Annotate the minor association (e.g., the observation of the
    association on few patients).
- Annotate the associations in the particular races.
- Annotate the association between disease and variant if the
    inheritance of the genetic disease is confirmed by the
variant.
- DO NOT annotate the symptoms of the target disease to the
    correlated variants or genes, unless the clear statement
between the symptom and variant is provided. (ex in PMID: 1952108)

### Gene-gene
- Annotate the pairs of the members in the protein complex, e.g.,
    EPO/EPOR (PMID:22808010).
    - In "We cloned the cDNA of three remaining human NADH:
        ubiquinone oxidoreductase subunits of this IP fraction: the
         NDUFS2 (49 kDa), NDUFS3 (30 kDa), and NDUFS6 (13 kDa)
        subunits.", all the pairs between two of the NDUFS2, NDUFS3
        , and NDUFS6 are annotated. (PMID:9647766)
- DO NOT annotate the "is_a" association between two genes. For
    instance, DO NOT annotate "breast and ovarian
cancer gene" and "BRCA1" (PMID:8944024).
- Annotate protein-protein interaction for singling proteins and
    the receptors. For instance, "Epo-R signaling
proteins (Akt, STAT5, p70s6k, LYN, and p38MAPK)" (PMID:27640183)

### Gene-chemical
- Annotate the chemical with the gene receptor. For instance, "
    antipsychotic drugs that have a high affinity with the
D2 receptor" (PMID:16867246)
```

```
### Chemical-chemical
- DO NOT annotate the "is_a" association between two chemicals.
   For instance, "Nefiracetam is a novel
pyrrolidone derivative" (PMID:8829135)
- Annotate the chemical-chemical association for the chemical
   contributes to the other for treatment, side effects and
drug-drug interaction. In "the midline serotonin B3 cells in the
   medulla contribute to the hypotensive action of
methyldopa", the pair of serotonin and methyldopa is annotated. (
   PMID:2422478)

### Chemical-variant
- Annotate the chemical with the variant, if the chemical binding
   ability of the gene impaired by the variant. For
instance, "This variant (apoE Guangzhou) may cause a marked
   molecular conformational change of the apoE and
thus impair its binding ability to lipids." (PMID:18046082)
- Annotate the chemical with the variant, if different mRNA
   stability between different alleles affected by the
chemical. For instance, "After transfection and inhibition of
   transcription with actinomycin D, analysis of mRNA
turnover failed to reveal differences in mRNA stability between
   A118 and G118 alleles, indicating a defect in
transcription or mRNA maturation." (PMID:16046395)

### Others
- DO NOT annotate the pairs of concepts with negative conclusions
   (e.g., "not significant", "no correlation was
observed"). For instance, "For most of the variants identified in
   the Kenyan and Sudanese study population, a
causative association with NSARD appears to be unlikely" is not
   annotated.
- In the case that a disease is induced by a chemical, and the
   other concepts (e.g., chemical, or protein) treats/affects
the induced disease, we annotate the three pairs by following
   examples. For instance:
   - "In addition, there is convincing clinical evidence that
      monotherapy with continuous subcutaneous apomorphine
      infusions is associated with marked reductions of
      preexisting levodopa-induced dyskinesias." (PMID:11009181)
      - Positive_Correlation between levodopa and dyskinesias
      - Negative_Correlation between apomorphine and dyskinesias
      - Negative_Correlation between apomorphine and levodopa
   - Absence of PKC-alpha attenuates lithium-induced nephrogenic
      diabetes insipidus. (PMID:25006961)
```

```
      - Positive_Correlation between lithiumand nephrogenic
          diabetes
      - Positive_Correlation between PKC-alpha and dyskinesias
      - Positive_Correlation between PKC-alpha and lithium
   - Characterization of a novel BCHE "silent" allele: point
      mutation (p.Val204Asp) causes loss of activity and
      prolonged apnea with suxamethonium. (PMID:25054547)
      - Positive_Correlation between p.Val204Asp and apnea
      - Negative_Correlation between apnea and suxamethonium
      - Association between p.Val204Asp and suxamethonium
      - Association between BCHE and apnea
      - Association between BCHE and suxamethonium
   - Curcumin prevents maleate-induced nephrotoxicity (PMID
      :25119790)
      - Negative_Correlation between Curcumin and nephrotoxicity
      - Positive_Correlation between maleate and nephrotoxicity
      - Negative_Correlation between Curcumin and maleate
- Annotate the previous reported association, even the conclusion
   demonstrates the association is not significant. (PMID
   :15824163)

### Guideline of the relation types
We collected a list of relation types between two concepts (e.g.,
   upregulation between two genes). Further, we merged some of
   the relation types to narrow down the curation complexity and
   increase the number of instances in each relation type.

The following mappings define how directional relation types
   should be converted to the BioRED relation types.
**Gene <-> Gene**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Upregulation | Positive_Correlation |
| Downregulation | Negative_Correlation |
| Regulation | Association |
| Positive_Correlation | Positive_Correlation |
| Negative_Correlation | Negative_Correlation |
| Bind | Bind |
| Modification | Association |
| Association | Association |

**Chemical <-> Gene**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Exhibition (Chemical -> Gene) | Positive_Correlation |
| Response (Gene -> Chemical) | Positive_Correlation |
```

```
| Suppression (Chemical -> Gene) | Negative_Correlation |
| Resistance (Gene -> Chemical) | Negative_Correlation |
| Association | Association |
| Receptor | Bind |
| Chem_Modification | Association |

**Chemical <-> Variant**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Association | Association |
| Resistance | Negative_Correlation |
| Response (and sensitivity) | Positive_Correlation |

**Gene <-> Disease**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Positive_Correlation | Positive_Correlation |
| Negative_Correlation | Negative_Correlation |
| Association | Association |

**Variant <-> Disease**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Cause | Cause |
| Association | Association |

**Chemical <-> Disease**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Treatment | Negative_Correlation |
| Induce | Positive_Correlation |
| Association | Association |

**Chemical <-> Chemical**
| Directional relation | BioRED relation |
|----------------------|-----------------|
| Cotreatment | Cotreatment |
| Inhibition | Negative_Correlation |
| Increase | Positive_Correlation |
| Drug_Interaction | Drug_Interaction |
| Association | Association |
| Comparison | Comparison |

### Disease-chemical
- Positive_Correlation
   - Chemical-induced disease.
```

```
   - Chemical (or Higher dose of the chemical) causes a higher
      risk of the disease.
   - Disease causes the increase of the chemical measured level.
   - The level of the chemical and the risk of the disease
      present a positive correlation.
   - Chemical exposures during development alter disease
      susceptibility later in life.
- Negative_Correlation
   - The disease-treated chemical/drug.
   - Disease causes the decrease of the chemical measured level.
   - The chemical/drug drops down the susceptibility of the
      disease.
- Association
   - A safety drug of the potential disease (PMID:20722491 -
      capecitabine(C110904) - hepatic and renal dysfunctions(
      D008107|D007674))
   - The associations of the pairs which cannot be categorized to
       positive/negative correlation.
   - The associations without clear description.

### Disease-gene
- If an association between a variant and a disease is confirmed,
   the corresponding gene should associate with the disease by "
   Association" type.
- Positive_Correlation
   - The overdose of protein causes the disease.
   - The knockout gene prevents the disease.
   - The side effect of protein (drug) causes the disease
- Negative_Correlation
   - The protein (drug) is used to treat/prevent the disease.
   - Lack of the protein causes the disease.
   - The knockout gene causes the disease.
- Association
   - The associations of the pairs which cannot be categorized to
       previous relation types.
   - The associations without clear description.
   - The functional gene prevents the occurrence of the disease.
   - Protein deficiency.

### Disease-variant
- Positive_Correlation
   - The variant increases the risk of the disease.
   - Significant frequency (p-value) of the disease with the
      specific allele.
   - The variant causes the gene to be either over-express or non-
      functional and further causes disease (or raises the
```

```
      disease susceptibility).
   - Disease is caused by protein deficiency, and the variant is
      responsible for the deficiency.
   - The variant plays a role in genetic predisposition to the
      disease.
   - The variant has a significant contribution to the disease.
   - Annotate "Positive_Correlation" to the pair of "founder
      mutation" and the target disease. (PMID:10788334,19394258)
- Negative_Correlation
   - The variant decreases the risk of the disease.
- Association
   - The variant observed from a number of patients.
   - The variant is associated with a lower prevalence of the
      disease or is responsible for the lower disease
      susceptibility
   - The associations of the pairs which cannot be categorized to
       "Cause" relation type.
   - The associations without clear description.
### Gene-gene
- Positive_Correlation
   - Two genes present the positive correlation in gene
      expression results.
   - Gene A is a transcription factor of the gene B, and gene A
      upper regulates the gene B.
   - Two genes present a positive correlation in any way.
- Negative_Correlation
   - Two genes present the negative correlation in gene
      expression results.
   - Gene A is a transcription factor of the gene B, and gene A
      down regulates the gene B.
   - Two genes present a negative correlation in any way.
- Bind
   - physical interaction between two proteins.
   - Protein A binds the promoter of gene B.
   - Two or more proteins in a complex.
   - Annotate "bind" to the protein and its protein receptor. ("
      androgen receptor" and "androgen" in PMID:15599941)
   - If the binding causes a positive or negative correlation,
      annotate the association to Positive_Correlation/
      Negative_Correlation.
   - DO NOT annotate the protein bind on a gene promoter region.
- Association
   - Annotate the modification (e.g, phosphorylation,
      dephosphorylation, acetylation ,deacetylation and other
      modifications) to association.
```

```
    - The associations of the pairs which cannot be categorized to
        any other association types.
- The associations without clear description.

### Gene-chemical
- If an association between a variant and a chemical is confirmed,
     the same association type should be
assigned to the corresponding gene of the variant and the chemical
    .
    - Positive_Correlation
    - The chemical causes a higher expression of the gene.
    - Higher gene expression causes higher sensitivity of the
       chemical.
    - The variant triggers the chemical adverse effects or causes
       the side effect worse.
    - The gene causes the chemical over-response.
    - Two genes present a positive correlation in any way.
- Negative_Correlation
    - The chemical causes a lower expression of the gene.
    - Higher gene expression causes lower sensitivity (resistance)
        of the chemical, and vise versa.
    - The variant has a protective role for the development of the
        chemical adverse effects.
    - The gene causes the chemical resistance.
    - Two genes present a negative correlation in any way.
- Association
    - The associations of the pairs which cannot be categorized to
        any other association types.
    - The associations without clear description.
- Bind
    - A chemical binds the promoter of a gene.
    - A protein is the chemical receptor.
    - If the binding causes a positive or negative correlation,
       annotate the association to Positive_Correlation/
       Negative_Correlation.

### Chemical-chemical
- Positive_Correlation (between A and B)
    - Chemical A increases the sensitivity of the chemical B.
    - Chemical A increases the treatment/inducing effectiveness of
        the chemical B to a disease.
    - Chemical A increases the effectiveness of the gene
       activation caused by chemical B.
- Negative_Correlation (between A and B)
    - Chemical A decreases the sensitivity of the chemical B.
```

```
    - Chemical A decreases the treatment/inducing effectiveness of
         the chemical B to a disease.(PMID:25080425 betaine
        attenuates isoproterenol-induced acute myocardial injury in
         rats.)
    - Chemical A decreases the side effects of the chemical B. (
        PMID:24587916 LF(D007781) - Dexamethasone(D003907))
    - Chemical A decreases the gene activation caused by chemical
        B. (PMID:17035713 Caspase activation by cisplatin was
        inhibited by CAA)
- Association
    - Annotate the chemical conversion to association type. (PMID
        :17391797 that catalyzes the conversion of
    phosphatidylethanolamine to phosphatidylcholine.)
    - The associations of the pairs which cannot be categorized to
         any other association types.
    - The associations without clear description.
- Drug_Interaction
- A pharmacodynamic interaction occurs when two chemicals/drugs
    are given together.
- Cotreatment
- Combination therapy.
- Conversion
- A chemical converse to the other chemical.

### Chemical-variant
- Positive_Correlation
    - The chemical causes a higher expression of the gene because
        of the specific variant.
    - The variant causes higher sensitivity of the chemical.
    - The variant causes the chemical over-response.
    - The gene and the chemical present a positive correlation in
        any way.
- Negative_Correlation
    - The chemical causes a lower expression of the gene because
        of the specific variant.
    - The variant causes lower sensitivity of the chemical.
    - The variant causes the chemical resistance.
    - The gene and the chemical present a negative correlation in
        any way.
    - The variant presents a protective role to the chemical
        adverse effect (or chemical caused disease).
- Association
    - The associations of the pairs which cannot be categorized to
         positive/negative correlation.
    - The associations without clear description.
```

```
- variant located on a chemical specific binding site, e.g.,
    the sequence variant c.465G>T encodes a conservative amino
    acid substitution, p.Glu155Asp, located in EF-hand 4, the
    calcium binding site of GCAP2 protein. (PMID:21405999)
```

Appendix B

Extraction Prompts

Note: **[MORE DEPTH]**

B.1 Prompt Version 1

```
You are a biomedical information extraction system.

Extract all biomedical entities and relationships from the
    following abstract.

Return ONLY valid JSON in the following form:

{
       "entities": [
              {
                     "text": "entity text",
                     "type": "entity type"
              }
       ],
       "relationships": [
              {
                     "source": "entity 1",
                     "relation": "relation type",
                     "target": "entity 2"
              }
       ]
}

Title:

{title}

Abstract:

{abstract}
```

B.2 Prompt Version 2

Baseline + BioRED annotation schema

```
You are an expert biomedical information extraction system.

Your task is to extract biomedical entities and relationships from
     the abstract below exactly as they would be annotated under
    the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

{biored_ext_guidelines}

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant

- OrganismTaxon

- CellLine

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
- Conversion
- Cotreatment
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

Return JSON in exactly this format:

{
       "entities": [
              {
                     "text": "entity text",
                     "type": "entity type"
              }
       ],
       "relationships": [
              {
```

```
                    "source": "entity 1",
                    "relation": "relationship type",
                    "target": "entity 2"
             }
      ]
}

Title:

{title}

Abstract:

{abstract}
```

B.3 Prompt Version 3

Baseline + BioRED annotation schema + few-shot prompting

```
You are an expert biomedical information extraction system.

Your task is to extract biomedical entities and relationships from
    the abstract below exactly as they would be annotated under
   the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

{biored_ext_guidelines}

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant

- OrganismTaxon

- CellLine

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
```

```
- Conversion
- Cotreatment
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

Below are examples of correctly annotated BioRED abstracts -
    follow the same annotation policy demonstrated in the examples.


{few_shot_block}

Return JSON in exactly this format:

{
	"entities": [
		{
			"text": "entity text",
			"type": "entity type"
		}
	],
	"relationships": [
		{
			"source": "entity 1",
			"relation": "relationship type",
			"target": "entity 2"
		}
	]
}

Title:

{title}

Abstract:

{abstract}
```

B.4 Prompt Version 4

Explicit exclusion of background biomedical knowledge, co-occurrence rule & explicit formatting

```
# TASK
```

```
You are an expert biomedical information extraction system.

Your task is to extract biomedical entities and relationships from the abstract below exactly as they would be annotated under the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

# BioRED Annotation Guidelines

{biored_ext_guidelines}

# Allowed Entity Types

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant

- OrganismTaxon

- CellLine

# Allowed Relationship Types

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
- Conversion
- Cotreatment
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

# Evidence requirements

- Base all extractions only on information explicitly supported by the provided abstract.
DO NOT infer entities or relationships from background biomedical knowledge.
- Entity co-occurence alone does not establish a relationship.
```

```
# Examples

Below are examples of correctly annotated BioRED abstracts -
    follow the same annotation policy demonstrated in the examples.


{few_shot_block}

# Output Format

Return JSON in exactly this format:

{
	"entities": [
		{
			"text": "entity text",
			"type": "entity type"
		}
	],
	"relationships": [
		{
			"source": "entity 1",
			"relation": "relationship type",
			"target": "entity 2"
		}
	]
}

# Title:

{title}

# Abstract

{abstract}
```

B.5 Prompt Version 5

Explicit prevention of unsupported specific relationship inference

```
# TASK

You are an expert biomedical information extraction system.
```

```
Your task is to extract biomedical entities and relationships from
     the abstract below exactly as they would be annotated under
    the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

# BioRED Annotation Guidelines

{biored_ext_guidelines}

# Allowed Entity Types

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant

- OrganismTaxon

- CellLine

# Allowed Relationship Types

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
- Conversion
- Cotreatment
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

# Evidence requirements

- Base all extractions only on information explicitly supported by
     the provided abstract.
- Do not infer a more specific relationship type than is
    explicitly supported by the abstract and BioRED annotation
    guidelines.
- In particular, do not classify an association as
    Positive_Correlation or Negative_Correlation unless the text
    provides evidence supporting that specific relationship type.
```

```
- DO NOT infer entities or relationships from background
    biomedical knowledge.
- Entity co-occurence alone does not establish a relationship.

# Examples

Below are examples of correctly annotated BioRED abstracts -
    follow the same annotation policy demonstrated in the examples.


{few_shot_block}

# Output Format

Return JSON in exactly this format:

{
	"entities": [
		{
			"text": "entity text",
			"type": "entity type"
		}
	],
	"relationships": [
		{
			"source": "entity 1",
			"relation": "relationship type",
			"target": "entity 2"
		}
	]
}

# Title:

{title}

# Abstract

{abstract}
```

B.6 Prompt Version 6

Stricter relation evidence requirements to reduce over-extraction.

```
# TASK
```

```
You are an expert biomedical information extraction system.

Your task is to extract biomedical entities and relationships from
     the abstract below exactly as they would be annotated under
    the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

# BioRED Annotation Guidelines

{biored_ext_guidelines}

# Allowed Entity Types

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant

- OrganismTaxon

- CellLine

# Allowed Relationship Types

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
- Conversion
- Cotreatment
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

# Evidence requirements

- Base all extractions only on information explicitly supported by
     the provided abstract.
Extract only entities and relationships that would be explicitly
    annotated under the BioRED annotation guidelines.
- Do not exhaustively generate relationships between entities
```

```
    merely because they are affected by the same intervention,
    participate in the same experimental context, or are discussed
     together.
- A relationship must correspond to a specific relationship
    expressed or clearly asserted in the abstract.
- Do not infer a more specific relationship type than is
    explicitly supported by the abstract and BioRED annotation
    guidelines.
- In particular, do not classify an association as
    Positive_Correlation or Negative_Correlation unless the text
    provides evidence supporting that specific relationship type.
- DO NOT infer entities or relationships from background
    biomedical knowledge.
- Entity co-occurence alone does not establish a relationship.

# Examples

Below are examples of correctly annotated BioRED abstracts -
    follow the same annotation policy demonstrated in the examples.


{few_shot_block}

# Output Format

Return JSON in exactly this format:

{
	"entities": [
		{
			"text": "entity text",
			"type": "entity type"
		}
	],
	"relationships": [
		{
			"source": "entity 1",
			"relation": "relationship type",
			"target": "entity 2"
		}
	]
}

# Title:

{title}
```

```
# Abstract

{abstract}
```

B.7 Prompt Version 7

Precision-focused conservative extraction to reduce over-extraction.

```
# TASK

You are an expert biomedical information extraction system.

Your task is to extract biomedical entities and relationships from
     the abstract below exactly as they would be annotated under
    the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

# BioRED Annotation Guidelines

{biored_ext_guidelines}

# Allowed Entity Types

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant

- OrganismTaxon

- CellLine

# Allowed Relationship Types

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
- Conversion
- Cotreatment
```

```
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

# Evidence requirements

- When uncertain whether an entity or relationship meets the
   BioRED annotation criteria, DO NOT extract it. Prioritise
   annotation precision over exhaustive extraction.
- Base all extractions only on information explicitly supported by
    the provided abstract.
- Extract only entities and relationships that would be annotated
   under the BioRED annotation guidelines.
- Do not exhaustively generate relationships between entities
   merely because they are affected by the same intervention,
   participate in the same experimental context, or are discussed
    together.
- A relationship must correspond to a specific relationship
   expressed or clearly asserted in the abstract.
- Do not infer a more specific relationship type than is supported
    by the abstract and BioRED annotation guidelines.
- In particular, do not classify an association as
   Positive_Correlation or Negative_Correlation unless the text
   provides evidence supporting that specific relationship type.
- DO NOT infer entities or relationships from background
   biomedical knowledge.
- Entity co-occurrence alone does not establish a relationship.

# Examples

Below are examples of correctly annotated BioRED abstracts -
   follow the same annotation policy demonstrated in the examples.


{few_shot_block}

# Output Format

Return JSON in exactly this format:

{
       "entities": [
              {
                     "text": "entity text",
                     "type": "entity type"
              }
```

```
        ],
        "relationships": [
                {
                        "source": "entity 1",
                        "relation": "relationship type",
                        "target": "entity 2"
                }
        ]
}

# Title:

{title}

# Abstract

{abstract}
```

B.8 Prompt Version 8

Targeted suppression of residual relation over-extraction through entity-pair evidence constraints

```
# TASK

You are an expert biomedical information extraction system.

Your task is to extract biomedical entities and relationships from
    the abstract below exactly as they would be annotated under
    the official BioRED annotation guidelines.

The official BioRED annotation guidelines are provided below.

# BioRED Annotation Guidelines

{biored_ext_guidelines}

# Allowed Entity Types

Output ONLY the following entity types:

- ChemicalEntity
- DiseaseOrPhenotypicFeature
- GeneOrGeneProduct
- SequenceVariant
```

```
- OrganismTaxon

- CellLine

# Allowed Relationship Types

Output ONLY the following relationship types:

- Association
- Bind
- Comparison
- Conversion
- Cotreatment
- Drug_Interaction
- Negative_Correlation
- Positive_Correlation

# Evidence requirements

- When uncertain whether an entity or relationship meets the
   BioRED annotation criteria, DO NOT extract it. Prioritise
   annotation precision over exhaustive extraction.
- Base all extractions only on information explicitly supported by
    the provided abstract.
- Extract only entities and relationships that would be annotated
   under the BioRED annotation guidelines.
- Do not exhaustively generate relationships between entities
   merely because they are affected by the same intervention,
   participate in the same experimental context, or are discussed
    together.
- A relationship must correspond to a specific relationship
   expressed or clearly asserted in the abstract.
- Do not infer a relationship merely because one entity is a
   symptom, characteristic, measurement, component, subtype,
   consequence, or contextual feature of another entity. Such
   semantic relatedness does not by itself constitute a BioRED
   relationship.
- Do not generate relationships simply because two entities are
   mentioned as part of the same disease, phenotype, treatment,
   pathway, experiment, comparison, or study outcome. The
   abstract must specifically support a BioRED relationship
   between those entities.
- Do not generate multiple relationships from a shared
   experimental result unless each individual entity pair is
   itself explicitly supported as a relationship under the BioRED
```

```
    annotation guidelines.
- Before extracting each relationship, verify that the abstract
   provides evidence for that specific entity pair and that the
   relationship satisfies the BioRED annotation criteria. If
   either condition is uncertain, omit the relationship.
- Do not infer a more specific relationship type than is supported
    by the abstract and BioRED annotation guidelines.
- In particular, do not classify an association as
   Positive_Correlation or Negative_Correlation unless the text
   provides evidence supporting that specific relationship type.
- DO NOT infer entities or relationships from background
   biomedical knowledge.
- Entity co-occurrence alone does not establish a relationship.

# Examples

Below are examples of correctly annotated BioRED abstracts -
   follow the same annotation policy demonstrated in the examples.


{few_shot_block}

# Output Format

Return JSON in exactly this format:

{
	"entities": [
		{
			"text": "entity text",
			"type": "entity type"
		}
	],
	"relationships": [
		{
			"source": "entity 1",
			"relation": "relationship type",
			"target": "entity 2"
		}
	]
}

# Title:

{title}
```

```
# Abstract

{abstract}
```

Appendix C

EPI Error Analysis

C.1 Error Categorisation Prompt

```
You are reviewing a relation-extraction error made by an NLP
    system.

Error type: {error_type}
- fp: a predicted relation that did not match the BioRED ground
    truth under the evaluation procedure.
- fn: a BioRED ground-truth relation that was not matched by a
    prediction.

Relation being analysed:
{error}

Abstract:{abstracts.set_index("pmid").loc[pmid]}

Ground-truth relations for this abstract:
{chr(10).join(str(gt) for gt in matching_gts)}

Categorise the primary cause of this error into exactly one of the
     categories below. Each category has a name and a description
    explaining when it applies - use the descriptions to guide
    your decision.

{category_list_str}

Use the abstract and BioRED ground truth as the reference when
    making your decision.
If multiple categories could apply, select the category that best
    explains why the prediction and ground truth did not match.

Respond with ONLY the category number, nothing else.
```

C.2 EPI Error Distribution Figures

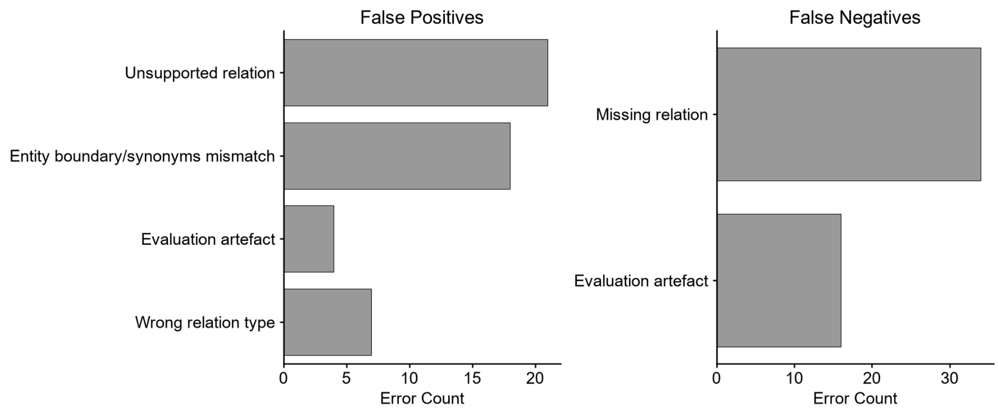

Figure C.1: `epi_003` relation error distribution.

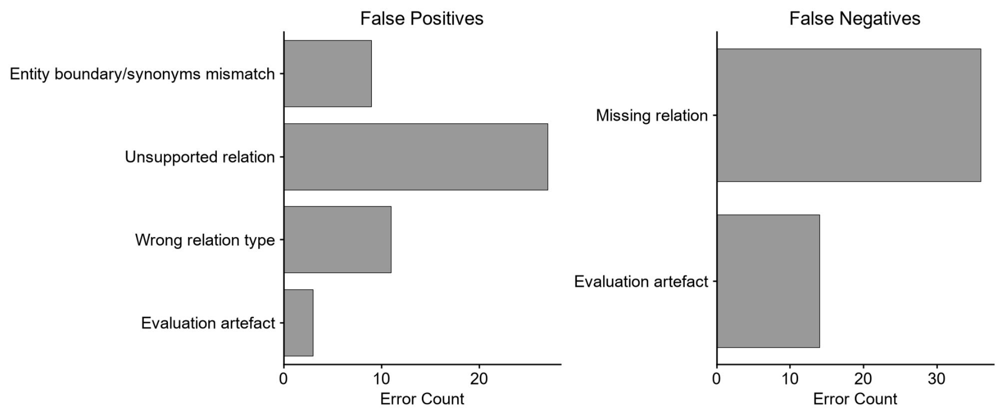

Figure C.2: `epi_006` relation error distribution.

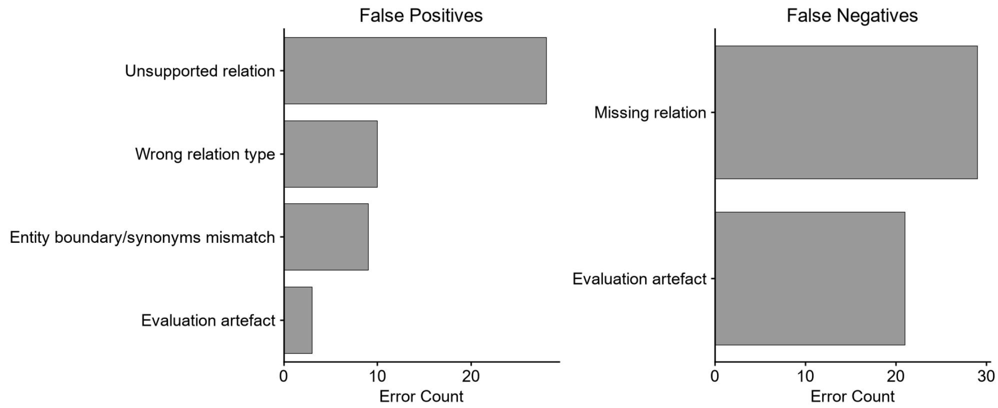

Figure C.3: `epi_009` relation error distribution.

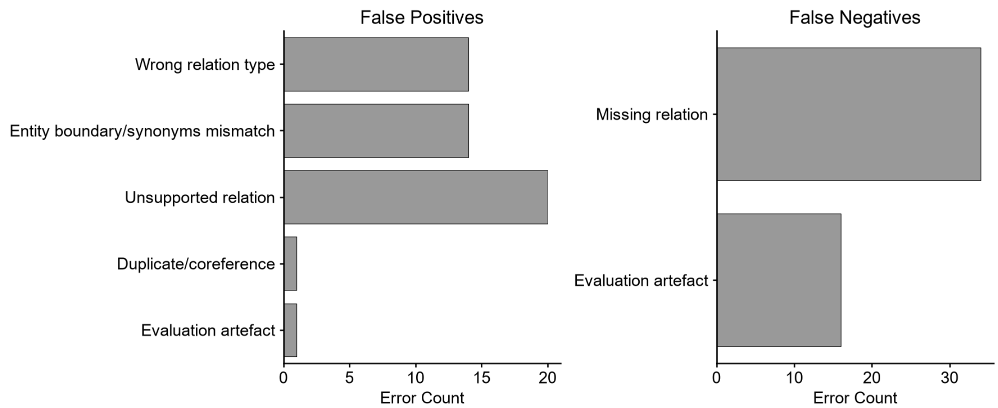

Figure C.4: `epi_011` relation error distribution.